The Goldbach–Chen Frontier: How the Semiprime Channel Inherits the Singular Series

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Abstract

The binary Goldbach conjecture asks whether every even $N > 2$ is a sum of two primes; Chen's theorem replaces the second prime by a prime or semiprime. We study how the Goldbach count $R_1(N)$ (prime q) and the Chen-surplus count $R_2(N)$ (semiprime q) respond to the Hardy–Littlewood singular series $\mathfrak{S}(N)$. Decomposing the semiprime channel $q = rs$ by its small prime factor r, each channel carries its own two-form singular series, and in the ratio R_2/R_1 the Goldbach singular series *cancels* (an unconditional Euler-product identity, Lemma 1), leaving a Mertens channel sum $\widetilde{W}(N) \sim \log\log N$. Thus $\mathfrak{S}(N)$ enters R_2 with exponent 1, exactly as in R_1: R_2 inherits the full singular series. The "half power" $R_2 \propto \mathfrak{S}^{1/2}$ reported by finite-range log–log fits is therefore not an exponent but a regression *slope* of a non-power-law object, obeying $\beta_2(N) = 1 - c(N)/\langle R_2/R_1\rangle \to 1$ with deficit $\sim 1/\log\log N$ (a multi-channel effect: the $3 \mid N$ atom accounts for only $\sim 10\%$ of it). A circle-method second-moment bound—using only Vaughan's classical *prime* minor-arc estimate and the trivial L^2 size of the semiprime sum, no sieve—gives the *linear* almost-all asymptotic $R_2(N) = \mathrm{main}(N)(1 + o(1))$ unconditionally in outline; the passage to the log-regression statement $\beta_2 \to 1$ is isolated as one explicitly stated lower-tail hypothesis rather than overclaimed. We complement this with reproducible diagnostics and a gallery of cross-disciplinary reframings, all governed by the same $\mathfrak{S}(N)$—one fact in many mirrors—of which two (an empirical circle method whose comet spectrum is the major arcs, and a primorial-base carry geometry) add a genuinely new computation. The goal is not a proof of Goldbach but a clean new identity, reproducible evidence, and a bounded reduction.

Contents

1 Introduction

The binary Goldbach conjecture states that every even integer $N > 2$ can be written as the sum of two primes. The conjecture remains open. However, several nearby facts are known. Helfgott proved the ternary, or weak, Goldbach conjecture: every odd integer greater than 5 is the sum of three primes [4]. Oliveira e Silva, Herzog and Pardi verified the binary Goldbach conjecture computationally up to $4 \cdot 10^{18}$ [9]. Chen's theorem proves that every sufficiently large even integer can be written as

$$N = p + q,$$

where p is prime and q is either prime or the product of at most two primes [1]; the semiprime channel and the "E_2" numbers it counts are classical sieve objects [2], and for the broader additive-bases context we refer to Nathanson [8]. Recent sieve-theoretic work also studies intermediate statements between Chen's theorem and the binary Goldbach conjecture [6].

This project focuses on the transition between the Goldbach representation

$$N = p + \text{prime}$$

and the Chen representation

$$N = p + \text{semiprime}.$$

The guiding question is not merely whether a representation exists, but how the available representations are distributed, how often the second term is genuinely prime, how often it is semiprime, and whether a representation can be selected in a smooth way as N varies.

Main idea. For each even N, define the number of representations with q prime and with q semiprime. Then study the empirical object

$$N \longmapsto \{(p,q) : p+q=N,\ p\in\mathbb{P},\ q\in\mathbb{P}\cup\mathcal{S}_2\}.$$

This is a discrete object, so it is not continuous in the ordinary analytic sense. Nevertheless, one can ask whether the induced empirical measures converge weakly, whether normalized counts have stable asymptotics, and whether a mixed-integer model can select one representation per even integer with small variation. This gives a precise mathematical language for the intuition that the representation may "jump" between prime and semiprime behavior.

Summary of findings. Sections 7–7.7 report the results of implementing this program; all numbers are reproducible from the accompanying code. The main outcomes are:

1. **The singular series of R_2, and the half-power mirage (§7.2).** Counts are computed for all $N \le 2\cdot 10^6$ by FFT and locally in windows up to 10^9 by a block-convolution sieve. Both R_1 and R_2 carry $\mathfrak{S}(N)$ with exponent 1: in the ratio R_2/R_1 the singular series *cancels* (Lemma 1), leaving the Mertens channel sum $\widetilde{W}(N) \sim \log\log N$. The number $\beta_2 \approx \frac{1}{2}$ that a finite-range log–log fit reports is therefore not an exponent but a regression slope of a non-power-law object; a derivation and the data agree on $\beta_2(N) = 1 - c/\langle R_2/R_1\rangle_N$ with $\langle R_2/R_1\rangle \sim \log\log N$, so $\beta_2 \to 1$ with an $\mathcal{O}(1/\log\log N)$ deficit (validation $(1-\beta_2)\langle R_2/R_1\rangle \approx$ 1.09 to 1%). The slope is multi-channel: deleting the heaviest channel ($3 \mid N$) removes only $\sim 10\%$ of it (§7.2), refuting a single-atom reading.
2. **Layers and sign reversal (§7.3).** For $\Omega(q)=k$, the slope β_k falls through $1, \frac{1}{2}, \ldots$ and becomes negative for $k \ge 3$: $\mathfrak{S}(N)$ enhances low-complexity and suppresses high-complexity channels.
3. **Diagnostics, continuity, optimization (§7.4–7.7).** Factor geometry favours unbalanced semiprimes; the q/N measures stabilize weakly to a near-uniform law; minimum-variation selection is a Pareto trade-off between geometric and type smoothness; the Chen-rescue threshold under a balance constraint is unimodal; and the residual balance exponent $a_{\mathrm{Res}}(N)$ correlates positively with $\mathfrak{S}(N)$.
4. **An almost-all theorem, honestly bounded (Appendix A).** The cancellation identity (Lemma 1) is unconditional; a circle-method second-moment bound—Vaughan's classical *prime* minor-arc estimate plus the trivial L^2 size of the semiprime sum, no sieve—gives the *linear* asymptotic $R_2(N) = \mathrm{main}(N)(1+o(1))$ for all but $o(X)$ even N, unconditional in outline. Upgrading this to the log-regression statement $\beta_2(N) \to 1$ needs one further input (lower-tail control of R_2/R_1, beyond L^2); we state that gap rather than claim past it.
5. **Many mirrors, one fact (§7.9–7.17).** Goldbach can be recast as a thermodynamic ground state, an adversarial network, a liquidity market, a control problem, an empirical circle method, an information channel, a discrete surface, and a carry geometry. We are explicit that this is *one* finding—$\mathfrak{S}(N)$ sets the local densities—observed many times, not many findings; most reframings are deterministic functions of the same $\mathfrak{S}/R_2$ data. Two earn their place by yielding a *new computation*: the empirical circle method (§7.14), whose comet power spectrum is literally the major arcs, and the carry geometry (§7.17), which surfaces a digit/Kummer signature of primality. The honest negatives (additive-complexity topology is empty; the Chen substitute carries no resilience premium) are reported as such.

2 Basic notation

Let $\mathbb{P}$ denote the set of primes. For an integer $n \geq 2$, let $\Omega(n)$ be the number of prime factors of n counted with multiplicity. Thus $\Omega(12) = 3$ because $12 = 2^2 \cdot 3$.

Definition 1 (Semiprimes and Chen numbers)**.** *Define*

$$\mathcal{S}_2 := \{n \geq 2 : \Omega(n) = 2\}$$

to be the set of semiprimes, allowing squares of primes. Define also

$$\mathcal{P}_2 := \{n \geq 2 : \Omega(n) \leq 2\} = \mathbb{P} \cup \mathcal{S}_2.$$

Thus $\mathcal{P}_2$ is the set of integers with at most two prime factors counted with multiplicity.

For an even integer N, define the prime–prime representation count

$$R_1(N) := \sum_{p<N} \mathbf{1}_{\mathbb{P}}(p)\mathbf{1}_{\mathbb{P}}(N-p). \tag{1}$$

This counts ordered representations by choosing the first summand p. If unordered counts are desired, one may restrict to $p \leq N/2$.

Define the prime–semiprime representation count

$$R_2(N) := \sum_{p<N} \mathbf{1}_{\mathbb{P}}(p)\mathbf{1}_{\mathcal{S}_2}(N-p). \tag{2}$$

Finally, define the Chen count

$$R_{\leq 2}(N) := R_1(N) + R_2(N) = \sum_{p<N} \mathbf{1}_{\mathbb{P}}(p)\mathbf{1}_{\mathcal{P}_2}(N-p). \tag{3}$$

Remark 1. *The binary Goldbach conjecture is equivalent to $R_1(N) > 0$ for every even $N > 2$. Chen's theorem implies $R_{\leq 2}(N) > 0$ for all sufficiently large even N.*

3 The Goldbach–Chen frontier

The central empirical object is the decomposition of Chen representations into genuinely Goldbach and genuinely semiprime representations:

$$R_{\leq 2}(N) = R_1(N) + R_2(N).$$

This suggests several diagnostics.

3.1 Chen surplus and Goldbach fragility

Definition 2 (Chen surplus)**.** *For even N, define the Chen surplus ratio*

$$C(N) := \frac{R_2(N)}{R_1(N) + 1}. \tag{4}$$

Large values of $C(N)$ identify integers for which semiprime representations are much more abundant than prime–prime representations.

Definition 3 (Goldbach fragility)**.** *For a threshold $k \in \mathbb{N}$, define the set of k-fragile even integers up to X by*

$$\mathcal{F}_k(X) := \{N \leq X : N \equiv 0 \pmod 2,\ R_1(N) \leq k,\ R_2(N) > 0\}. \tag{5}$$

These are integers for which Goldbach representations exist only sparsely, while Chen-type representations remain available.

Research question. How large is $\mathcal{F}_k(X)$? Are fragile integers concentrated in specific congruence classes? Does fragility correlate with the singular series correction in the Hardy–Littlewood heuristic?

3.2 Prime versus semiprime share

Define

$$\theta(N) := \frac{R_1(N)}{R_1(N) + R_2(N)}, \tag{6}$$

with the convention $\theta(N) = 0$ if the denominator is zero.

Here $\theta(N)$ is the empirical share of Chen representations in which q is prime rather than semiprime. A small value of $\theta(N)$ means that most Chen representations use a genuinely semiprime second summand.

A naive density heuristic suggests

$$R_1(N) \asymp \frac{N}{\log^2 N}, \qquad R_2(N) \asymp \frac{N \log\log N}{\log^2 N},$$

because primes have local density roughly $1/\log N$, while semiprimes have density roughly $\log\log N/\log N$. Therefore, at a first heuristic level, semiprime representations should outnumber prime–prime representations by a factor of order $\log\log N$. This does not prove Goldbach or Chen, but it motivates the following empirical conjecture.

Conjecture 1 (Average semiprime dominance)**.** *There exists a slowly varying normalization* $L(X) \sim \log\log X$ *such that*

$$\frac{1}{X/2} \sum_{\substack{N \le X \\ N \equiv 0\,(2)}} \frac{R_2(N)}{R_1(N) + 1} \asymp L(X).$$

Equivalently, prime–semiprime representations should become increasingly abundant relative to prime–prime representations on average.

4 Distribution of the second summand q

For each even N, define the representation sets

$$\mathcal{R}_1(N) := \{(p,q) : p + q = N,\ p \in \mathbb{P},\ q \in \mathbb{P}\}, \tag{7}$$

$$\mathcal{R}_2(N) := \{(p,q) : p + q = N,\ p \in \mathbb{P},\ q \in \mathcal{S}_2\}, \tag{8}$$

$$\mathcal{R}_{\le 2}(N) := \mathcal{R}_1(N) \cup \mathcal{R}_2(N). \tag{9}$$

The object $q = N - p$ can be studied in several ways.

4.1 Location of q inside $[0, N]$

Define empirical measures on $[0, 1]$:

$$\mu_N^{(1)} := \frac{1}{R_1(N)} \sum_{(p,q) \in \mathcal{R}_1(N)} \delta_{q/N}, \tag{10}$$

$$\mu_N^{(2)} := \frac{1}{R_2(N)} \sum_{(p,q) \in \mathcal{R}_2(N)} \delta_{q/N}, \tag{11}$$

$$\mu_N^{(\le 2)} := \frac{1}{R_{\le 2}(N)} \sum_{(p,q) \in \mathcal{R}_{\le 2}(N)} \delta_{q/N}, \tag{12}$$

whenever the denominators are nonzero.

Question 1 (Weak continuity). *Do the measures* $\mu_N^{(1)}$, $\mu_N^{(2)}$, *or* $\mu_N^{(\le 2)}$ *converge weakly along typical even* N*? If so, is the limit close to the uniform distribution on* $(0,1)$ *after removing small endpoint effects?*

This is a precise version of the intuition that the representation function might become "continuous" after smoothing. The pointwise object is discontinuous, because primality is discrete; but the normalized cloud of representations may stabilize in distribution.

4.2 Type-labeled empirical measures

Let the type of q be

$$\tau(q) := \begin{cases} 1, & q \in \mathbb{P}, \\ 2, & q \in \mathcal{S}_2. \end{cases}$$

Define a type-labeled empirical measure on $[0,1] \times \{1,2\}$ by

$$\nu_N := \frac{1}{R_{\le 2}(N)} \sum_{(p,q)\in\mathcal{R}_{\le 2}(N)} \delta_{(q/N,\tau(q))}. \tag{13}$$

Conjecture 2 (Weak type stabilization). *For typical large even* N*, the measures* ν_N *are close to a product-like law*

$$\nu_N \approx f_N(x,t)\,dx\,d\tau,$$

where the spatial component in $x = q/N$ *varies slowly, while the type marginal increasingly favors* $t = 2$ *because semiprimes have larger density than primes.*

This is not ordinary continuity. It is closer to convergence of empirical distributions, or continuity after smoothing over windows $N \in [X, X+H]$.

4.3 Factor geometry of semiprime q

If $q \in \mathcal{S}_2$, write

$$q = ab, \qquad a, b \in \mathbb{P}, \qquad a \le b.$$

Then define the balance statistic

$$B(q) := \frac{\log a}{\log q} \in \left(0, \frac{1}{2}\right]. \tag{14}$$

A value near $1/2$ means that q is a balanced semiprime; a value near 0 means that q has a very small prime factor.

Question 2 (Semiprime factor geometry). *Among representations* $N = p + q$ *with* $q \in \mathcal{S}_2$*, how is* $B(q)$ *distributed? Are Chen representations dominated by unbalanced semiprimes with a small factor, or by balanced semiprimes?*

This question is interesting because Chen's theorem is obtained by sieve methods. Empirically, one can ask whether the extra freedom in q comes mostly from allowing small prime divisors, or whether semiprime representations are broadly distributed across factor shapes.

5 Heuristic asymptotics (classical recovery)

The asymptotics in this section are classical; we recall them only to fix normalisation and as a control for the computations of §7, not as new derivations. The Hardy–Littlewood circle-method heuristic for Goldbach [3] predicts a main term of the form

$$R_1(N) \approx 2C_2\,\mathfrak{S}(N)\frac{N}{\log^2 N}, \tag{15}$$

where C_2 is the twin-prime constant and $\mathfrak{S}(N) = \prod_{p|N,\, p>2} \frac{p-1}{p-2}$ is the singular-series correction depending on the odd prime divisors of N (the factor $2C_2$ already absorbs the $p = 2$ local density).

For semiprime q, Landau's density of numbers with two prime factors [5], $\#\{n \le x : \Omega(n) = 2\} \sim x \log\log x / \log x$, gives the rough estimate

$$\#\{n \le x : \Omega(n) = 2\} \sim \frac{x \log\log x}{\log x}. \tag{16}$$

Therefore, a first-order independence heuristic gives

$$R_2(N) \approx K_2(N) \frac{N \log\log N}{\log^2 N}, \tag{17}$$

where $K_2(N)$ is a correction factor capturing congruence constraints and local arithmetic structure.

Normalization. The empirical project should therefore compare

$$A_1(N) := \frac{R_1(N)}{N/\log^2 N}, \tag{18}$$

$$A_2(N) := \frac{R_2(N)}{N \log\log N/\log^2 N}. \tag{19}$$

If the heuristic is meaningful, $A_1(N)$ and $A_2(N)$ should be much more stable than the raw counts.

6 Selection formulations

Scope note. This section records the optimization *language* used by two diagnostics measured in §7.5 (a minimum-variation selection and a Chen-rescue threshold). It is deliberately short: the feasibility version is trivial and no MIP solver will prove Goldbach (Remark 2). The infinite-dimensional and column-generation readings are recorded in Appendix B; a reader after the empirical results may skip to §7.

Fix an even $N \le X$ and let $\mathcal{A}_N^{(1)} = \{p \in \mathbb{P} : N - p \in \mathbb{P}\}$ and $\mathcal{A}_N^{(2)} = \{p \in \mathbb{P} : N - p \in \mathcal{S}_2\}$. Binary variables $z_{N,p,t}$ select one representation $N = p + q$ of type t (prime $t = 1$, semiprime $t = 2$); the cover constraint $\sum_{p\in\mathcal{A}_N^{(1)}} z_{N,p,1} + \sum_{p\in\mathcal{A}_N^{(2)}} z_{N,p,2} \ge 1$ for every even N in a window $\mathcal{E}_X$ is Chen feasibility, and dropping the type-2 terms is Goldbach feasibility. With the sets $\mathcal{A}_N^{(t)}$ precomputed by a sieve this is trivial as a feasibility problem; it becomes informative only under added structure.

Remark 2 (Why this is not a proof strategy)**.** *These mixed-integer programs do not make Goldbach easier by themselves. They translate the conjecture into a covering problem over a hypergraph whose vertices are even integers and whose hyperedges are prime-pair certificates; the value is conceptual, letting one import selection, smoothness, total variation, and column generation into the empirical study of Goldbach-type representations (Appendix B).*

Minimum-variation selection. Selecting one representation per even N in a window, with type $u_N = \sum_{p\in\mathcal{A}_N^{(2)}} z_{N,p,2} \in \{0,1\}$ and normalized location $s_N = q_N/N = \sum_{p,t} \frac{N-p}{N} z_{N,p,t}$ (linear in the selection variables), the minimum-variation MIP minimizes $\alpha \sum_N v_N + \beta \sum_N r_N + \gamma \sum_N |s_N - \frac{1}{2}|$ subject to one selection per N and $v_N \ge |u_N - u_{N-2}|$, $r_N \ge |s_N - s_{N-2}|$. The term $\sum v_N$ measures type oscillation, $\sum r_N$ geometric oscillation, and $|s_N - \frac{1}{2}|$ rewards balance. The guiding question is whether the minimum type-switch rate $|\mathcal{E}_X|^{-1} \min \sum_N v_N$ decreases or stays bounded away from 0 as $X \to \infty$; §7.5 reports the answer (a smooth-location branch that nonetheless keeps switching type).

Chen-rescue threshold. Restricting representations to a balanced band $p \in [N/2 - k, N/2]$ and counting how many N *require* a semiprime (no prime pair in the band) defines the rescue fraction, whose unimodal profile in k is reported in §7.5; if the minimum number of semiprime selections is zero over a finite range, Goldbach holds there.

7 Empirical results and a new regularity

This section reports the outcome of implementing the program above. All numbers are reproducible from the accompanying code (see the Data and code availability statement; the drivers `run_counts.py`, `run_scaling.py`, `run_betalaw.py`, `run_estimator_audit.py`); unless stated otherwise they use the range $X = 2 \cdot 10^6$.

7.1 Computational setup and validation

The counts R_1 and R_2 are obtained for *all* $N \le X$ at once as additive convolutions,

$$R_1 = \mathbf{1}_{\mathbb{P}} * \mathbf{1}_{\mathbb{P}}, \qquad R_2 = \mathbf{1}_{\mathbb{P}} * \mathbf{1}_{\mathcal{S}_2},$$

evaluated by FFT in $O(X \log X)$. For $X = 2 \cdot 10^6$ this takes about 15 seconds including sieves, diagnostics and figures. The FFT counts agree exactly with a brute-force enumeration on random samples (the rounding error of the transform is far below 1/2), and Goldbach's condition $R_1(N) > 0$ is verified throughout the range.

7.2 The semiprime channel and the half-power mirage

The Hardy–Littlewood prediction (15) attaches to R_1 the full singular series $\mathfrak{S}(N) = \prod_{p|N,\, p>2}(p-1)/(p-2)$. Regressing $\log(R_1/(N/\log^2 N))$ against $\log \mathfrak{S}(N)$ recovers, empirically, a slope 1.000, and dividing by $\mathfrak{S}(N)$ collapses the coefficient of variation of the normalized count from 0.39 to 0.015 (a 25× reduction); this is a control that reproduces (15) and the band structure of the Goldbach comet.

The new observation concerns R_2. We measure how strongly the relative semiprime share tracks $\mathfrak{S}(N)$ by the *regression slope*

$$\beta_2 - 1 := \text{slope of} \quad \log \frac{R_2(N)}{R_1(N)} \quad \text{against} \quad \log \mathfrak{S}(N),$$

which uses R_1 itself as a meter for $\mathfrak{S}(N)$ and is therefore independent of any chosen normalization. We call β_2 the *empirical singular exponent*, but stress at the outset that β_2 **is a regression slope, not an exponent of a power law**: as Lemma 1 shows below, $\mathfrak{S}(N)$ enters R_2 with exponent 1 (exactly as in R_1) and *cancels* in the ratio R_2/R_1, so β_2 is the slope of a non-power-law object and the value $\approx \frac{1}{2}$ reported in finite ranges is a finite-size correction, not a structural exponent (developed after Table 1). We adopt one estimator throughout: ordinary least squares of $\log(R_2/R_1)$ on $\log \mathfrak{S}$, taken *cumulatively* over all even $5000 \le N \le X$. Table 1 reports it across several ranges; standard errors are $\le 10^{-4}$ (so the digits shown are significant), and the appendix (§A.6) instead uses a *local* window centred at X, which runs ≈ 0.02–0.03 higher at the same X—the distinction is between an average over $[5000, X]$ and a measurement at the leading edge.

We thus find the effective *finite-range* law (valid as a fit near $X \sim 10^6$, with the understanding that the exponent $\frac{1}{2}$ drifts to 1; §7.2 below)

$$R_2(N) \;\propto\; \mathfrak{S}(N)^{1/2} \frac{N}{\log^2 N} W(N), \qquad W(N) = \sum_{\substack{3 \le r \le \sqrt{N} \\ r \text{ prime}}} \frac{r-1}{r-2} \frac{1}{r} \frac{\log N}{\log(N/r)}, \tag{20}$$

X	β_2	R^2 of the regression
$5 \cdot 10^5$	0.452	0.91
$1 \cdot 10^6$	0.475	0.91
$2 \cdot 10^6$	0.497	0.92
$4 \cdot 10^6$	0.513	0.93
$8 \cdot 10^6$	0.529	0.93

Table 1: Cumulative regression slope β_2 of $\log(R_2/R_1)$ on $\log \mathfrak{S}$ over $[5000, X]$ (standard error $\leq 10^{-4}$). The value sits near $\frac{1}{2}$ in this range and *rises* with X—already a warning that it is not a fixed exponent (contrast $\beta_1 = 1.000$, which is genuinely scale-invariant). The singular series explains about 92% of the variance of $\log(R_2/R_1)$.

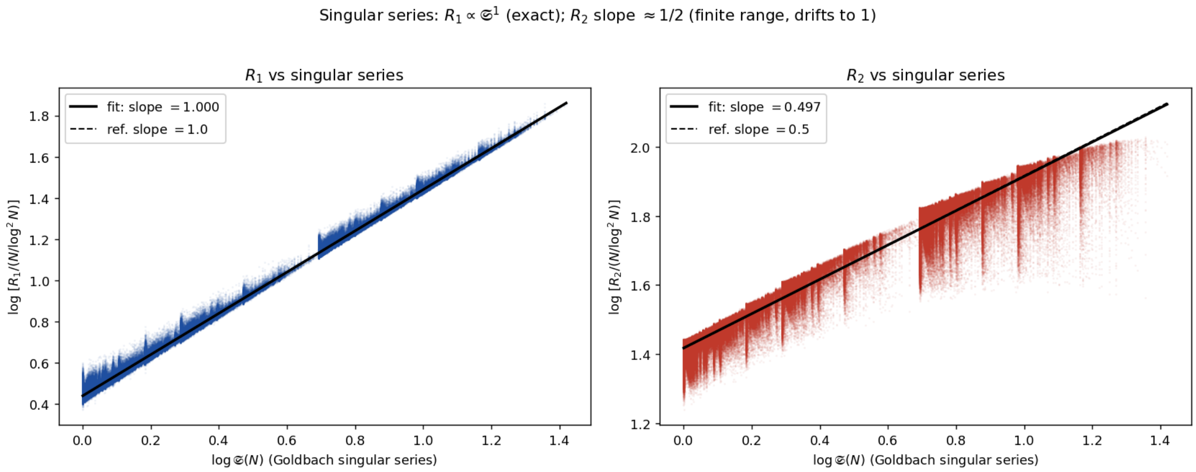

Figure 1: Left: $\log[R_1/(N/\log^2 N)]$ against $\log \mathfrak{S}(N)$ has slope 1.000 (Hardy–Littlewood). Right: the same for R_2 has slope 0.497 at $X = 2 \cdot 10^6$, near the reference line $\beta = 1/2$—but this is the *finite-range* slope of a non-power-law object (it drifts toward 1; Fig. 2), not a fixed exponent. The vertical striations are the discrete values of $\mathfrak{S}(N)$.

where $W(N) \sim \log\log N$ by Mertens' theorem [13]. The weight W makes precise the $\log\log N$ factor of (17): it is a Mertens sum over the *small prime factor* r of the semiprime $q = rs$, weighted by the twin-prime-type local factor $(r-1)/(r-2)$. Table 2 compares the dispersion of R_2 under three normalizations.

Interpretation: a quantitative Chen rescue effect. Because the finite-range slope has $\beta_2 < 1$, over this range the relative semiprime share falls with the singular series, $R_2/R_1 \propto \mathfrak{S}(N)^{-(1-\beta_2)}$ (locally $\approx \mathfrak{S}(N)^{-1/2}$): where Goldbach is locally rich (large $\mathfrak{S}$), the prime channel dominates, whereas where Goldbach is *fragile* (small $\mathfrak{S}$) the semiprime channel is proportionally most abundant. This makes the Chen-rescue intuition—the semiprime channel matters most exactly where primes are scarce—quantitative as an *effective finite-window scaling*, with the deficit $1 - \beta_2 \asymp 1/\log\log N$ vanishing asymptotically (so it is a slope, not a fixed power; §7.2 opening). Heuristically the slope sits below 1 because writing $q = rs$ couples the divisors of N to the factor r: when a small prime $\ell \mid N$, the channel $r = \ell$ is *blocked* (then $p = N - \ell s \equiv 0 \bmod \ell$ forces $p = \ell$), removing its term from the sum over channels precisely when $\mathfrak{S}(N)$ is large (Lemma 1). This explains the direction $\beta_2 < 1$; the precise value is not 1/2, as the scaling analysis below shows.

normalization of R_2		CV $(X = 2 \cdot 10^6)$
$N \log\log N/\log^2 N$	(heuristic of §5)	0.184
$2C_2\,\mathfrak{S}(N)^1\,N\,W/\log^2 N$	(full inheritance)	0.174
$2C_2\,\mathfrak{S}(N)^{1/2}\,N\,W/\log^2 N$	(**half power**)	**0.018**

Table 2: At $X = 2 \cdot 10^6$ the half-power normalization fits about $10\times$ tighter than the $\log\log N$ heuristic, and tighter than the full singular series ($\beta = 1$)—but this is a statement about the *finite-range* value of the slope, not about the limiting exponent. As X grows the best-fitting β rises toward 1 (Fig. 2); the local optimality of $\beta = \frac{1}{2}$ here reflects the $1/\log\log N$ deficit at $X \sim 10^6$, not a failure of R_2 to inherit $\mathfrak{S}(N)$.

Scaling to 10^9: the slope grows like $\log\log X$. Measuring β_2 *locally* in a window of $2 \cdot 10^6$ even integers centred at X (computed by a block-convolution sieve, so that X can reach 10^9 while only the window is held in memory), the slope does *not* settle at $1/2$: it rises monotonically (Fig. 2). Eight windows from $6 \cdot 10^6$ to 10^9 are fit to within 0.0013 by

$$\beta_2(X) \;\approx\; -0.094(7) + 0.234(3)\,\log\log X \qquad (R^2 = 0.9987,\ 8 \text{ windows}), \tag{21}$$

the parenthesised figures being the last-digit standard errors of the two-parameter fit. Thus the half-power $\beta_2 \approx \frac{1}{2}$ is only the *local* value near $X \sim 10^6$, and the slope is not a constant. The fit reaches $\beta_2 \approx 0.61$ at 10^9 (measured). We deliberately *do not* read a numerical limit off (21): extrapolating a $\log\log X$ trend fitted over three decades to, say, $X \sim 10^{46}$ (where it would touch $\beta_2 = 1$) is well outside the data and is not evidence—the robust content is the *functional form* $1 - \beta_2 \asymp 1/\log\log X$ and its derivation below, not a forecast of where the slope crosses a particular value. The regression quality stays high throughout ($R^2 \approx 0.997$, CV of the half-power normalization $\lesssim 0.04$).

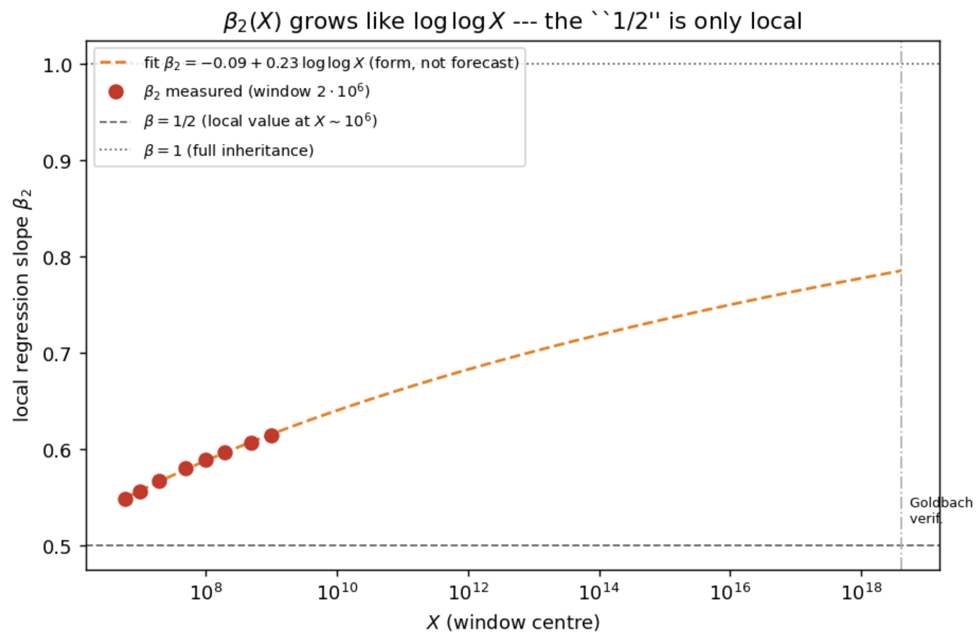

Figure 2: Local regression slope $\beta_2(X)$ for windows up to 10^9, with the $\log\log X$ fit (21) (the dashed extrapolation beyond 10^9 is shown only to display the functional form, not as a prediction). The slope drifts steadily above $1/2$ without sign of leveling off, consistent with $\beta_2 \to 1$.

A derivation: $\beta_2 \to 1$ with an $1/\log\log \mathbf{N}$ deficit. The drift is not arbitrary. Writing $q = N - p = rs$ and summing over the small factor r, each channel is a Goldbach count for the linear form $(s, N - rs)$, whose two-form singular series factorizes as

$$\mathfrak{S}_2(r, N) = \begin{cases} 2C_2\, \mathfrak{S}(N)\, \dfrac{r-1}{r-2}, & r \nmid N, \\ 0, & r \mid N \end{cases} \tag{22}$$

(Lemma 1; the $r \mid N$ channel is identically blocked). Since $R_1 = 2C_2\mathfrak{S}(N)I_1$ and $R_2 = 2C_2\mathfrak{S}(N)\sum_{r\nmid N} \frac{r-1}{r-2} I_r$, the ratio *cancels the singular series exactly*,

$$\frac{R_2(N)}{R_1(N)} = \widetilde{W}(N) := \sum_{\substack{3 \le r \le \sqrt{N} \\ r \nmid N}} \frac{r-1}{r-2}\, w_r(N), \qquad w_r = \frac{1}{r}\frac{\log N}{\log(N/r)}. \tag{23}$$

Hence the measured exponent is a regression slope, $\beta_2 - 1 = \operatorname{Cov}(\log \widetilde{W}, \log \mathfrak{S}) / \operatorname{Var}(\log \mathfrak{S})$, and the only coupling to $\mathfrak{S}(N)$ comes from the blocked channels $r \mid N$, each *removed* from $\widetilde{W}$. Modelling $\mathbf{1}[\ell \mid N]$ as independent with probability $1/\ell$, $\operatorname{Var}(\log \mathfrak{S}) =: B$ is a convergent constant, while the removed terms contribute $\operatorname{Cov}(\log \widetilde{W}, \log \mathfrak{S}) = -A/\widetilde{W}_0$ with $A > 0$ constant and $\widetilde{W}_0 \approx \langle R_2/R_1 \rangle \sim \log\log N$ the baseline sum (Theorem 1). Therefore

$$\beta_2(N) = 1 - \frac{c(N)}{\langle R_2/R_1 \rangle_N} + o\!\left(\frac{1}{\log\log N}\right), \qquad c(N) = \tfrac{A}{B} = O(1) \ (\approx 1.1 \text{ in range; Fig. 21}), \tag{24}$$

so $\beta_2 \to 1$: R_2 *asymptotically inherits the full Goldbach singular series*, the approach being governed by $1/\log\log N$. This explains why finite ranges show $\beta_2 \approx \frac{1}{2}$ and a slow apparent drift.

Proposition 1 (Validation, non-circular)**.** *The law* (24) *predicts that* $(1-\beta_2)\langle R_2/R_1 \rangle$ *is constant. Measured over windows from* $6 \cdot 10^6$ *to* $2 \cdot 10^8$ *it equals* $1.11, 1.11, 1.10, 1.09, 1.09, 1.08$*, a coefficient of variation of* 1% *(versus* 1.7% *for* $(1-\beta_2)\log\log N$*): the correct "clock" is* $\langle R_2/R_1 \rangle$*, exactly as* (24) *requires (Fig. 3).*

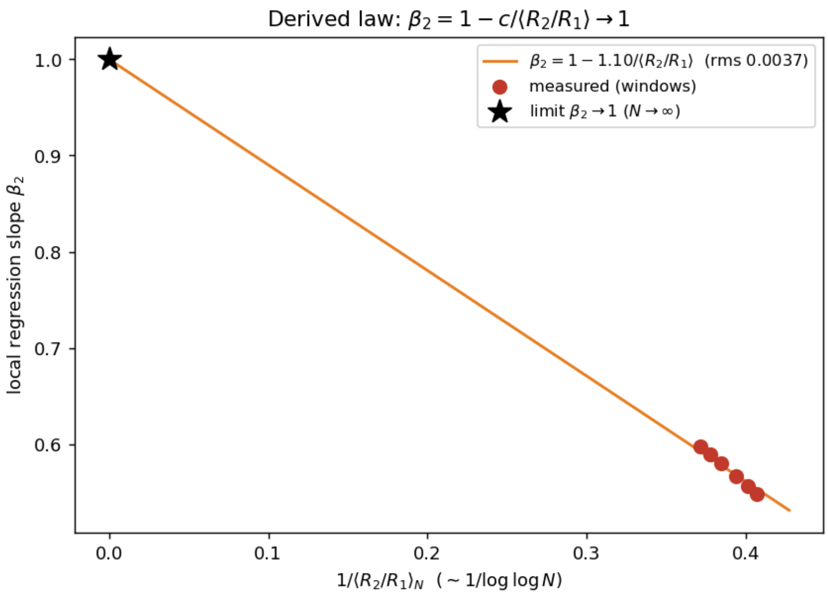

Figure 3: The derived law: β_2 against $1/\langle R_2/R_1 \rangle \sim 1/\log\log N$ is a straight line through the limit point $(0, 1)$, i.e. $\beta_2 \to 1$.

Not a single-atom effect. A natural worry is that the whole slope is the arithmetic of one prime: $3 \mid N$ simultaneously doubles $\mathfrak{S}(N)$ (the factor $\frac{3-1}{3-2} = 2$) and deletes the heaviest semiprime channel $r = 3$, so one might suspect $\beta_2 \approx \frac{1}{2}$ is just "what divisibility by 3 does" near $X \sim 10^6$. The data refute this. Recomputing the slope on the subpopulation $3 \nmid N$ (the $r = 3$ atom removed) gives, at $X = 2 \cdot 10^6$, $\beta_2 = 0.547 \pm 0.0004$ against 0.495 on all N: the deficit $1 - \beta_2$ falls by only about 10% (`run_estimator_audit.py`). The mechanism is real—conditional means confirm $\langle \log \mathfrak{S} \mid 3 \mid N\rangle - \langle \log \mathfrak{S} \mid 3 \nmid N\rangle = 0.69 \approx \log 2$ and a depressed R_2/R_1 on $3 \mid N$—but it accounts for a tenth of the slope, not the bulk. The negativity of $\beta_2 - 1$ is genuinely a *sum over channels* $r = 3, 5, 7, \ldots$, each blocked on the sparse set $r \mid N$, exactly as $A(X) = \sum_r (\cdots)$ in (24); it is not the signature of a single residue class.

7.3 The singular exponent across arithmetic-complexity layers

The decomposition R_1, R_2 extends to layers $R_k(N) = \#\{p < N : \Omega(N - p) = k\}$, counting representations whose second summand has exactly k prime factors (Goldbach is $k = 1$, Chen's semiprime channel is $k = 2$). Measuring the singular exponent β_k of each layer – the slope of $\log(R_k/R_1)$ against $\log \mathfrak{S}(N)$, plus one – reveals a clean pattern (Table 3, Fig. 4):

k	1	2	3	4	5	6
share $\rho(k)$	15.5%	33.5%	28.7%	14.6%	5.5%	1.8%
β_k	1.00	0.50	-0.30	-1.38	-2.73	-4.04

Table 3: Layer profile and singular exponent at $X = 2 \cdot 10^6$ (each regression has $R^2 \approx 0.92$). The exponent decreases monotonically and crosses zero between $k = 2$ and $k = 3$; it is *not* $1/k$.

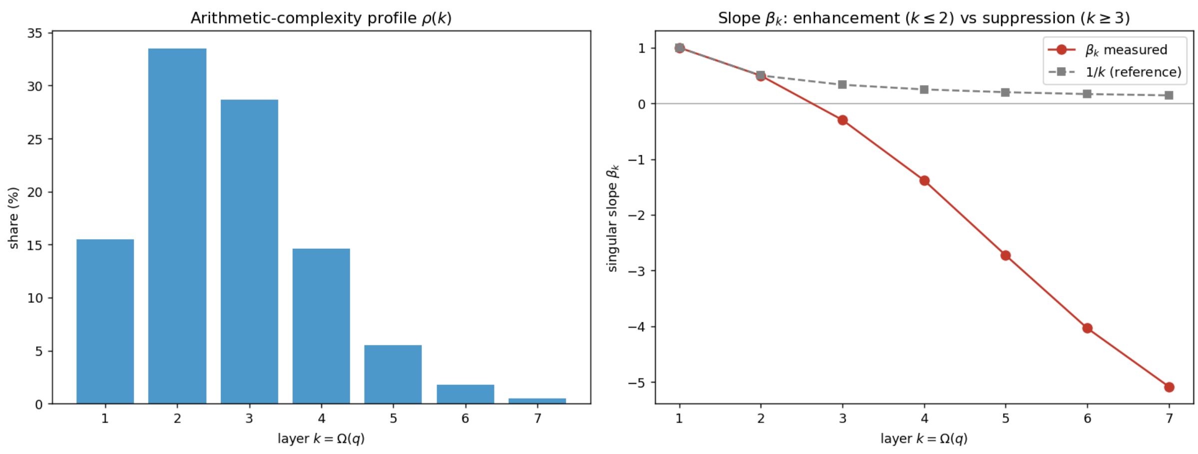

Figure 4: Left: the arithmetic-complexity profile $\rho(k)$ peaks at $k = 2$. Right: the singular exponent β_k decreases through $1, \frac{1}{2}, \ldots$ and becomes negative for $k \geq 3$, diverging from the naive $1/k$ reference.

Sign reversal of the singular response. The Goldbach singular series *enhances* the low-complexity channels ($\beta_1 = 1$, $\beta_2 \approx \frac{1}{2} > 0$) but *suppresses* the high-complexity ones ($\beta_k < 0$ for $k \geq 3$): where $\mathfrak{S}(N)$ is large, representations with a many-factored q become rarer. The mechanism is the same divisor effect seen for R_2, now run in reverse. When $\mathfrak{S}(N)$ is large, N is divisible by many small primes; the admissible p avoid those residues, which forces $q = N - p$ to also avoid small prime factors, so q tends to have *fewer* prime factors. Thus a large singular series biases q toward low arithmetic complexity. All β_k drift upward with X at the same slow rate as

β_2 (e.g. β_3 moves from -0.30 at $2 \cdot 10^6$ to -0.20 at $8 \cdot 10^6$), so the crossing point of $\beta_k = 0$ moves slowly to the right. We also find that the strongest "rescue" channel, $q =$ composite$\times$composite (both factors composite), is available for *every* even N in the range with on average $\sim 1.8 \cdot 10^4$ representations: the additive hierarchy is extremely redundant below the prime layer.

7.4 Diagnostics

The prime share has mean $\bar{\theta} = 0.311$ and decreases slowly with N (the semiprime channel gains ground, as the $\log \log N$ factor predicts); the Chen surplus has mean $\overline{C} = 2.27$. Goldbach fragility is extremely rare: $|\mathcal{F}_1| = 1$, $|\mathcal{F}_2| = 3$, $|\mathcal{F}_3| = 5$, $|\mathcal{F}_5| = 14$, $|\mathcal{F}_{10}| = 49$ up to X; and *every* fragile integer observed has $R_2 > 0$, so Chen always rescues. The semiprime factor balance $B(q) = \log a / \log q$ has weighted mean 0.25, with 16% of the representation mass at $B < 0.1$ (a tiny prime factor) and 18% at $B > 0.4$ (near balanced): Chen representations are dominated by *unbalanced* semiprimes with a small prime factor (Fig. 5), answering the semiprime factor-geometry question of §4.

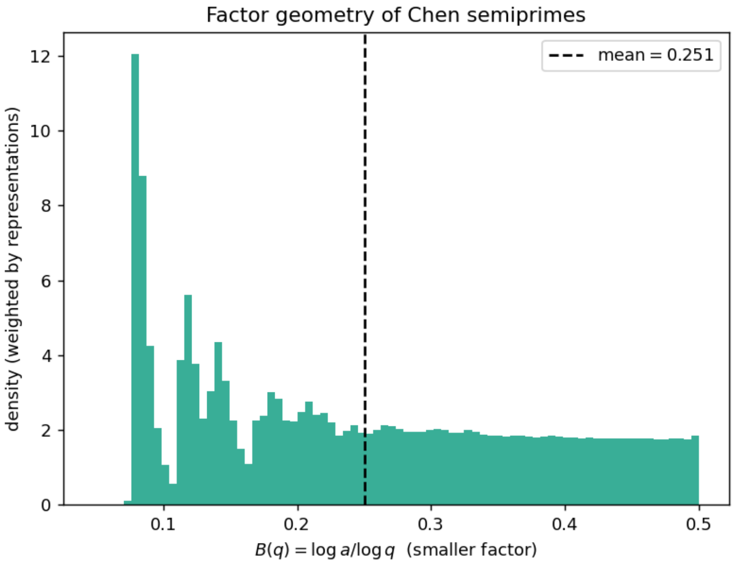

Figure 5: Representation-weighted distribution of the factor balance $B(q)$ of the semiprime second summand. Mass concentrates toward small B (small prime factor), not toward the balanced value 1/2.

7.5 Optimization experiments

Smoothness is a Pareto trade-off. Minimizing the total spatial variation $\sum |s_N - s_{N-2}|$ of the selected location $s_N = q_N / N$ yields a geometrically very smooth branch (total variation 0.0040 over 201 consecutive even integers) that nonetheless *switches type in* 28% *of its steps* and uses semiprimes 72% of the time (Fig. 6). Forbidding semiprimes (the pure Goldbach branch, with zero type switches) raises the variation to 0.0246, a factor of 6.2 rougher. One cannot simultaneously maximize geometric and type smoothness: continuity in location and chaos in type coexist.

A unimodal Chen-rescue threshold. Restricting representations to a balanced band $p \in [N/2 - k, N/2]$ and measuring the fraction of N that *require* a semiprime (no prime pair in the band) gives a unimodal curve in k (Fig. 7). For $k \gtrsim 0.005\,N$ the prime channel essentially never fails; the rescue fraction peaks near 38–40% at an intermediate half-width

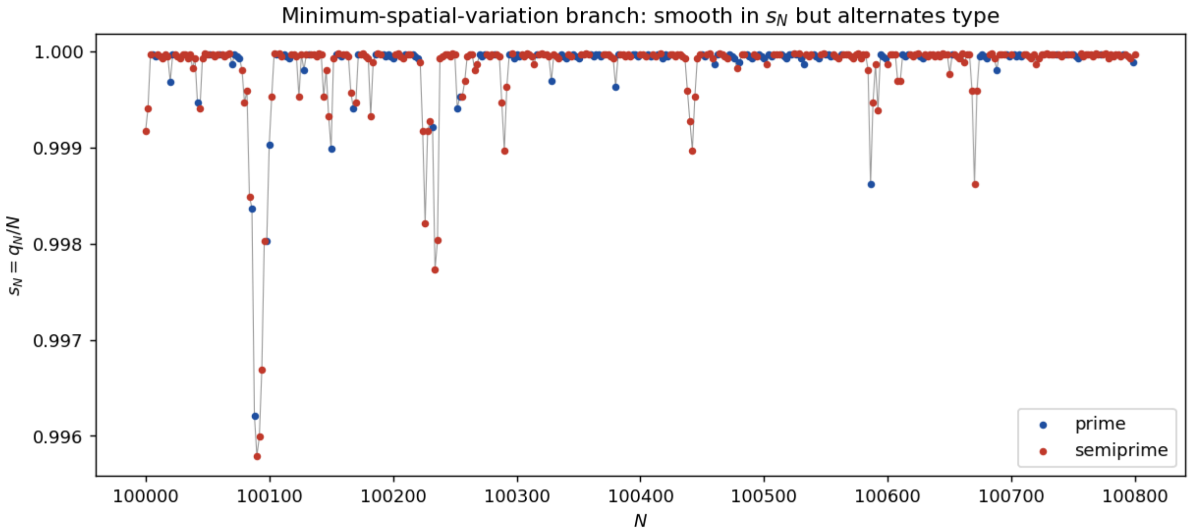

Figure 6: Minimum-variation branch (Gurobi). The optimum hugs $s_N \approx 1$ (the "least partner" branch, p a small prime) and is smooth in s_N, yet its type (blue prime / red semiprime) alternates.

$k \sim 20$–40; and for $k = 1, 2$ it falls again because often no representation exists at all. The peak shifts right with N (roughly as $\log^2 N$), marking where the expected number of balanced prime pairs, $\sim 2C_2\,\mathfrak{S}(N)\,k/\log^2(N/2)$, drops to $O(1)$ – a Poisson-type transition. This is the regime captured by the residual exponent $a_{\mathrm{Res}}(N)$ of §7.7: Chen's semiprime flexibility becomes *necessary* precisely when near-perfect balance is imposed.

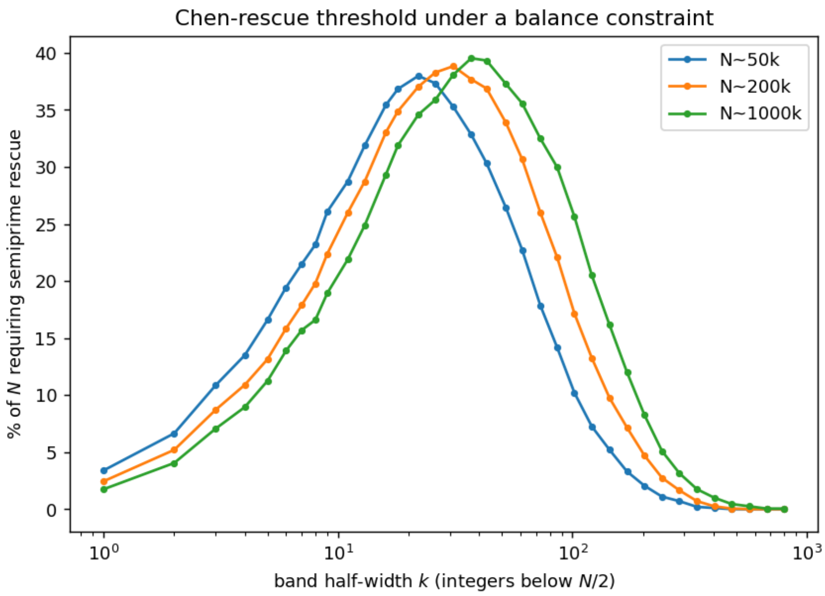

Figure 7: Fraction of N that require a semiprime rescue as a function of the band half-width k, for three scales of N. The curve is unimodal and its peak scales like $\log^2 N$.

7.6 Weak continuity of the location measures

Finally we test the weak-continuity intuition of §4 (Question 1 and the weak type-stabilization hypothesis there) directly. The pointwise measure μ_N of q/N is noisy, but averaging over a window $N \in [X, X + H]$ produces a stable density (Fig. 8, left). Three findings, at $X = 10^6$:

- **The smoothed measure converges.** Splitting a window into two same-scale subsamples, the total-variation distance between their densities falls monotonically from $2.1 \cdot 10^{-3}$ to $3 \cdot 10^{-4}$ as H grows from 125 to 4000 (Fig. 8, right), i.e. the per-N noise averages out – a concrete form of weak convergence.

- **The limit is essentially uniform.** The prime-q density is within total variation 0.028 of the uniform law on $(0, 1)$, with only a mild upturn near the endpoints (the $1/(\log(xN) \log((1 - x)N))$ effect). This answers Question 1 affirmatively: after removing endpoint effects, $\mu_N^{(1)}$ is close to uniform.

- **Prime and semiprime locations nearly coincide.** The two densities are within total variation 0.019 of each other (Fig. 8, center), confirming the weak type-stabilization picture. The semiprime location is slightly skewed toward larger q (mean $q/N = 0.513$ versus the exactly symmetric 0.500 for primes), the only visible signature that p and q play asymmetric roles in R_2.

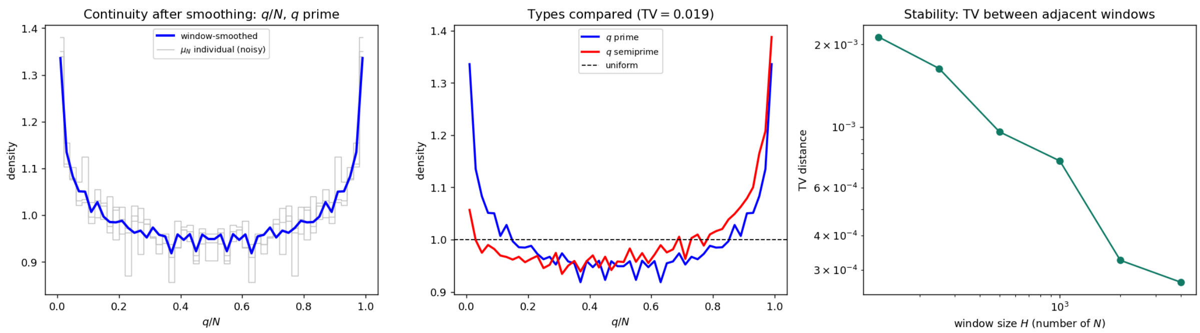

Figure 8: Weak continuity of q/N. Left: noisy pointwise μ_N (gray) versus the window-smoothed density (blue). Center: prime versus semiprime location densities, both close to uniform. Right: same-scale sampling distance decreases with the window size H (log–log).

7.7 The residual Goldbach–Chen balance exponent

Inspired by the $(1 + a)$ interpolation of Li and Liu (every large even N is $p + rq$ with $r \le q^{a-1}$, so that $(1+1)$ is Goldbach and $(1+2)$ is Chen), one can ask not whether a *fixed* exponent works, but the smallest exponent *needed* for each N. For the nondegenerate Chen layer $N = p + rs$ (p, r, s prime, $r \le s$), define the residual balance exponent and minimal multiplier

$$a_{\mathrm{Res}}(N) = 1 + \min_{N=p+rs} \frac{\log r}{\log s}, \qquad r_\star(N) = \min\{r \in \mathbb{P} : \exists p, s \in \mathbb{P},\ N = p + rs\}.$$

We compute both exactly for all even $N \le 2 \cdot 10^6$ ($r_\star$ by FFT channel coverage; a_{Res} by a small-p scan, validated against full enumeration to 0 error). Findings (Fig. 9, $N \ge 1000$):

- **Typical residual collapse.** The median a_{Res} *decreases* with X ($1.115 \to 1.099 \to 1.088$ at $X = 10^5, 5 \cdot 10^5, 2 \cdot 10^6$), with $p_{99} = 1.182$ and a finite-range maximum of 1.385 – far below the Li–Liu threshold 1.9. The worst case is pinned by a few small "hard" N.

- **The minimal multiplier is tiny.** $r_\star = 3$ for 68% of N and the dictionary $\{2, 3, 5, 7\}$ already covers 99.7%; $\{2, \ldots, 11\}$ covers 99.99%. This settles the multiplier-dictionary question empirically.

- **Local obstruction.** $r_\star$ jumps when a small prime divides N: if $3 \mid N$ the channel $r = 3$ is blocked (then $p = N - 3s \equiv 0 \bmod 3$), and indeed the mean $r_\star$ is 4.59 for $3 \mid N$ versus 2.94 otherwise.

- **Link to the singular series.** a_{Res} correlates *positively* with $\log \mathfrak{S}(N)$ (correlation $+0.59$). The very divisibility of N by small primes that enlarges $\mathfrak{S}(N)$ (boosting Goldbach) also blocks the small-r Chen channels, so the residual balance is *worst exactly where Goldbach is locally richest.* This ties the balance exponent to the singular-series theme of §7.3.

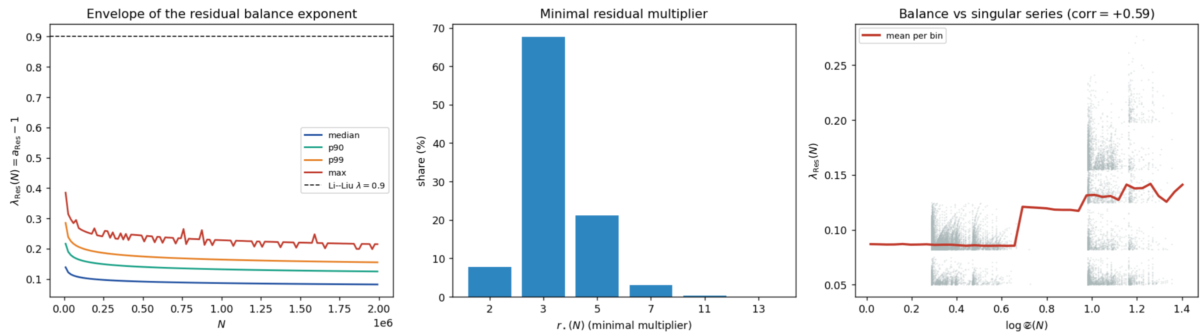

Figure 9: Left: percentile envelope of $\lambda_{\mathrm{Res}} = a_{\mathrm{Res}} - 1$ versus N, far below the Li–Liu line 0.9. Centre: the minimal multiplier $r_\star$ ($r = 3$ dominates). Right: λ_{Res} rises with the Goldbach singular series $\mathfrak{S}(N)$ (positive correlation).

7.8 Three further probes: transport, topology, dynamics

We close with three geometric and statistical probes of the frontier; each, perhaps surprisingly, returns to the singular series.

Optimal transport (reflection defect). Under the reflection $T(x) = 1 - x$ (which sends p/N to q/N), compare the prime location law α (density of p/N) with the reflected layer laws $T_\#\beta_k$. The reflection defects in W_1 are $\Delta_1 = W_1(\alpha, T_\#\beta_1) = 0.040$ and $\Delta_2 = W_1(\alpha, T_\#\beta_2) = 0.032$, and the optimal Chen mixture attains $\Delta_{\le 2} = 0.032$ at $\lambda^\star = 0$: the reflected *semiprime* cloud matches the primes *better* than the reflected primes do, a 22% "transport collapse" (stable in X), driven by unbalanced semiprimes with a small factor (Fig. 10).

Topology of valleys. Normalising each count by its own singular factor ($R_1/\mathfrak{S}$, $R_{\le 2}/(\mathfrak{S} + \mathfrak{S}^{1/2}W)$) and computing the 0-dimensional sublevel persistence of the detrended sequences, the valleys of the Goldbach and Chen comets largely *coincide*: the anomalies correlate at $+0.64$, and at the 200 most fragile N the Chen anomaly (0.29) is as deep as the Goldbach one (0.30). Chen does *not* topologically fill the Goldbach valleys (only the deepest 30% are softened, Fig. 11); the rescue is one of magnitude, not of valley structure, because both channels share the constraint that p be prime.

Statistical dynamics of the frontier. The prime share is almost a deterministic function of the singular series: $\theta(N)$ is 94% explained by $\mathfrak{S}(N)$ alone (adding $N \bmod 30$ contributes nothing further), and the increment $\Delta\theta_N$ is 98% explained by the jump $\mathfrak{S}(N) \to \mathfrak{S}(N+2)$. The residual is small (25% of the spread of θ) and is *not* white noise (short-range lag-1 autocorrelation -0.21 after detrending): the Goldbach–Chen frontier carries little irreducible randomness, being governed by the local arithmetic of N (Fig. 12).

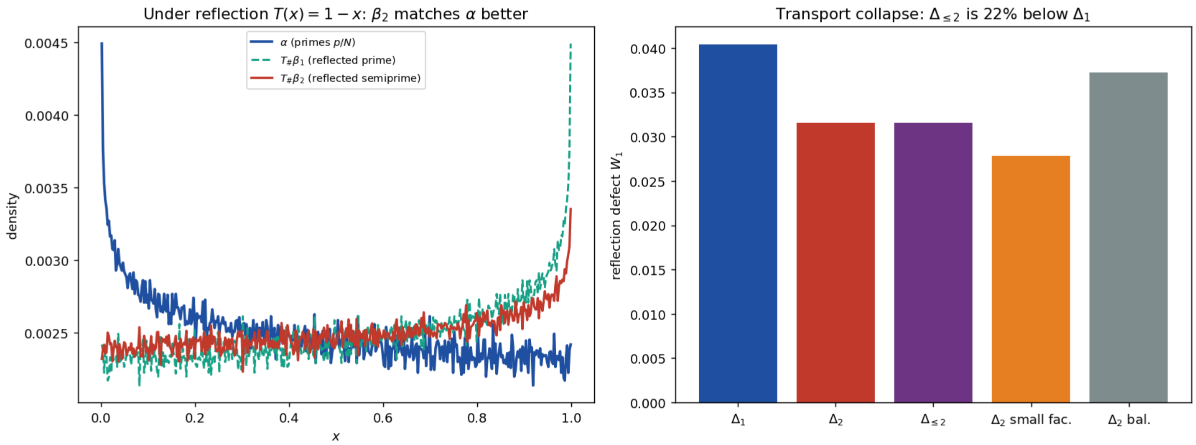

Figure 10: Reflection transport. Left: prime law α and the reflected prime and semiprime laws. Right: admitting semiprimes lowers the reflection defect by 22%.

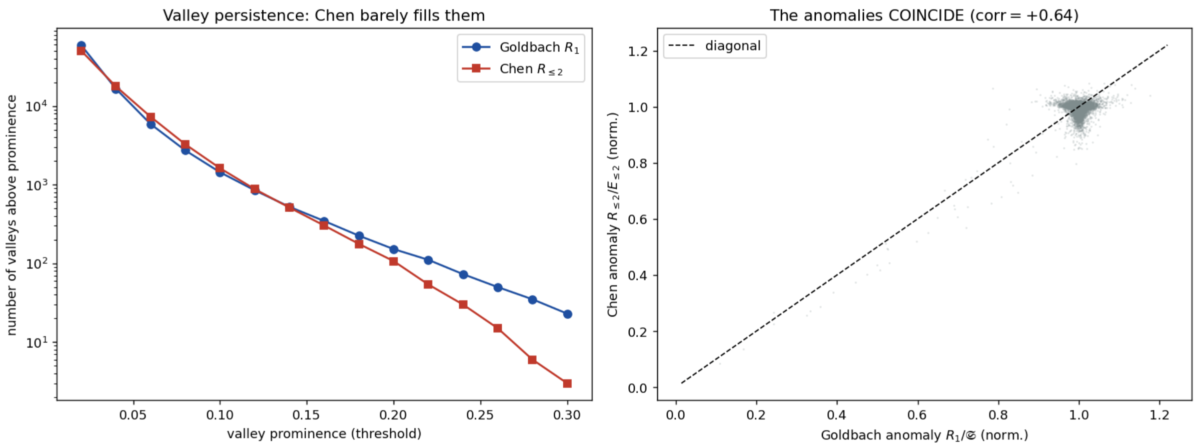

Figure 11: Valley persistence. Left: Chen has nearly as many valleys as Goldbach (slightly fewer deep ones). Right: the normalised anomalies coincide (+0.64).

A note on the reframings that follow

The subsections §7.9–7.17 recast the Goldbach–Chen frontier in the languages of statistical physics, network science, markets, control, the circle method, information theory, and digit geometry. We are explicit that these are *not* independent findings: in every case the governing quantity turns out to be the same singular series $\mathfrak{S}(N)$, so most of them are one fact—$\mathfrak{S}$ sets the local densities—re-encoded, and several (the melting temperature, the substitute spread, the information budget) are deterministic functions of the same $\mathfrak{S}/R_2$ data already plotted above. We keep them because the vocabularies are individually illuminating, but we flag, for each, whether it merely *re-describes* or actually *predicts or computes* something the bare law $R_2 \propto \mathfrak{S}$ does not. By that test two stand out and should be read as the genuine contributions of this section: the empirical circle method (§7.14), whose comet power spectrum literally exhibits the major arcs and quantifies how modulus-6 structure dominates, and the carry geometry (§7.17), which surfaces a new object—a digit-sum/Kummer-carry signature of primality. The control reframing (§7.13) is kept for its concrete search-heuristic benchmark; the topology probe (§7.12) is a reported negative. The reader interested only in the theorem may skip to the conclusion.

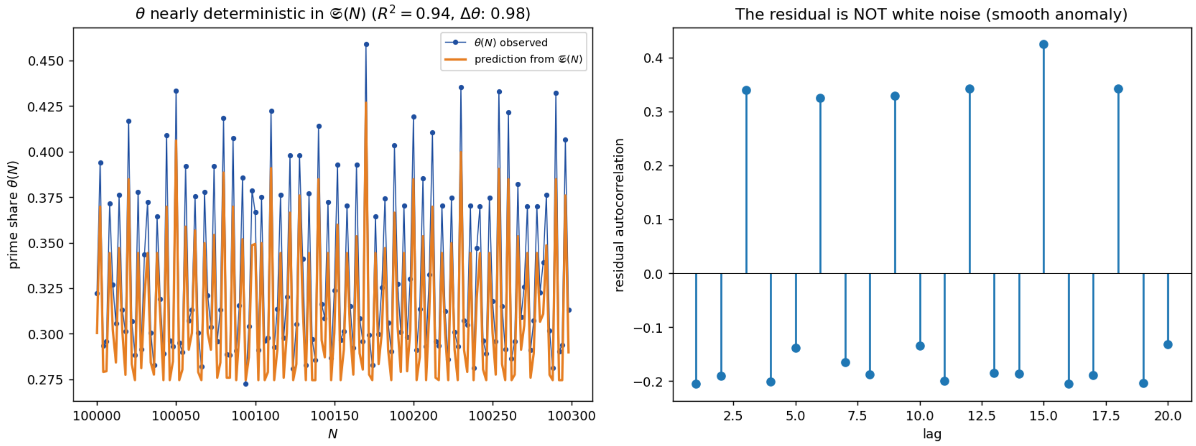

Figure 12: Dynamics. Left: $\theta(N)$ tracks its singular-series prediction. Right: the residual autocorrelation, small and non-white.

7.9 Goldbach as arithmetic ground-state energy

A single reformulation subsumes the counts R_1, R_2, the layers $\Omega(q) = k$, and the partition-function viewpoint. Assign to each decomposition $a + b = N$ $(a, b \geq 2)$ the *arithmetic energy* $H(a, b) = \Omega(a) + \Omega(b)$, and define the ground-state energy and level spectrum

$$E(N) = \min_{a+b=N} H(a, b), \qquad n_E(N) = \#\{a + b = N : \ \Omega(a) + \Omega(b) = E\}.$$

Then $E(N) = 2$ *iff* Goldbach holds for N; the ground-state degeneracy is $n_2(N) = R_1(N)$, and the first excited level is $n_3(N) = 2R_2(N)$ (prime+semiprime, Chen). The spectrum (Fig. 13, left) is a hump peaked at the typical energy $\langle H \rangle \sim 2 \log \log N$, with the Goldbach ground state sitting in its low-energy tail; the gap $\langle H \rangle - E(N) \sim 2 \log \log N - 2$ diverges.

The thermodynamics is computable for all N at once, since the partition function is a self-convolution,

$$Z_N(\beta) = \sum_{a+b=N} e^{-\beta H(a,b)} = (f_\beta * f_\beta)(N), \qquad f_\beta(n) = e^{-\beta\Omega(n)},$$

so one FFT per β yields Z_N, the mean energy $U_N(\beta) = -\partial_\beta \log Z_N$, and the specific heat $C_N(\beta) = \beta^2 \operatorname{Var}_\beta(H)$. As β increases past a "melting" point $\beta^\star(N)$, the system localises onto the Goldbach ground state: U_N drops from $\sim 2 \log \log N$ to 2 and $C_N(\beta)$ peaks at $\beta^\star$ (Fig. 13, centre; mean $\beta^\star \approx 2.1$). Strikingly, this melting temperature is almost a deterministic function of the singular series: $\operatorname{corr}(\beta^\star, \log \mathfrak{S}) = -0.92$ (Fig. 13, right). Where $\mathfrak{S}(N)$ is large the ground state is highly degenerate ($R_1 \propto \mathfrak{S}$) and entropically favoured, so Goldbach order melts at a *lower* β. Thus the same singular series that governs R_1, β_2 and the balance exponent also governs the thermodynamic stability of the Goldbach ground state—a unified statistical-mechanical picture in which Goldbach is the statement "every even N has ground-state energy 2".

7.10 Adversarial robustness of the Goldbach graph

View the representations as a bipartite graph: even integers N on one side, primes p on the other, with an edge $N \sim p$ when $N - p$ is prime; the degree of N is $R_1(N)/2$. How robust is Goldbach to deletion of primes? On a window of $1.5 \cdot 10^4$ even integers near 10^6 we compare three adversaries (Fig. 14).

Random deletion is almost powerless: deleting each prime independently with probability ϕ leaves N representable unless all its $\sim R_1(N)/2$ disjoint pairs are hit, an event of probability

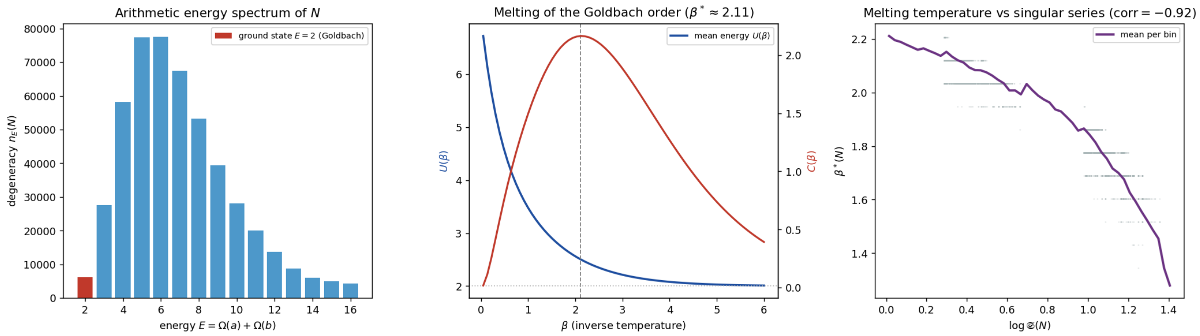

Figure 13: Arithmetic ground-state energy. Left: the energy spectrum $n_E(N)$, a hump at $\sim 2\log\log N$ with the Goldbach ground state $E = 2$ (red) in its tail. Centre: mean energy $U(\beta) \to 2$ and specific heat $C(\beta)$, whose peak $\beta^\star \approx 2.1$ is the melting temperature of Goldbach order. Right: $\beta^\star$ versus $\log \mathfrak{S}(N)$ (correlation -0.92).

$(1-(1-\phi)^2)^{R_1(N)/2}$. Goldbach survives in the *whole* window until $\phi \approx 0.97$; one may delete 95% of all primes at random and still represent every N.

Targeted deletion is efficient but local: since the pairs $\{p, N-p\}$ of a fixed N are disjoint, breaking N is a vertex cover of a matching, costing exactly $R_1(N)/2$ primes; the cheapest target is the lowest-degree N, giving $\kappa = \min_N R_1(N)/2$. In the window this is 4.6% of the primes—an order of magnitude less than the random threshold, but aimed at a single N.

Modular deletion is the striking case. Delete the primes in one reduced class $a \bmod m$. The survivors lie in $(\mathbb{Z}/m)^\times \setminus \{a\}$, and an N is destroyed when its class is missing from the survivors' sumset. For $m = 3$, deleting the class 1 (i.e. *half* of all primes) leaves only the class 2, whose sumset is $\{2+2\} = \{1\}$: every $N \equiv 0, 2 \pmod 3$ loses all its representations, destroying 59% of the window (Fig. 14, right). The same 50% of primes removed at random destroys nothing. Thus the adversary manufactures a *local obstruction*—the exact mechanism the singular series rules out for the primes themselves. The effect is confined to the smallest moduli: $m = 4$ deleting class 1 kills $N \equiv 0 \bmod 4$ (50%), while for $m \geq 5$ the surviving classes already sum onto everything and no N is fully obstructed.

In short, the Goldbach graph is *random-robust but structure-fragile*: invulnerable to losing almost all primes by chance, yet shattered by removing one arithmetic progression. This is a network-science counterpart of the local–global theme: Goldbach's redundancy is enormous, but it is arithmetic, not statistical, redundancy.

7.11 Goldbach as a market: liquidity and the fair-weather substitute

Read each even N as a market that must clear by hiring two prime inputs $p + q = N$; the liquidity is the number of matches, $L_G(N) = R_1(N)$. Chen lets one input be a *minimally composite substitute* (a semiprime), giving $L_C(N) = R_1(N) + R_2(N)$ and a substitute spread $S_C(N) = R_2/R_1 \approx 2.3$: in normal times the Chen market is three times more liquid. We ask whether the substitute also provides *resilience* under a supply shock that removes each prime with probability ϕ. A Goldbach match survives with probability $(1-\phi)^2$ (two primes), but a Chen match only with $(1-\phi)^3$, because the semiprime $q = rs$ needs its two factors r, s to survive as well. Hence the expected surviving liquidities are $L_G(\phi) = R_1(1-\phi)^2$ and $L_C(\phi) = L_G(\phi) + R_2(1-\phi)^3$, and the substitute spread *decays linearly*,

$$S_C(\phi) = \frac{L_C(\phi) - L_G(\phi)}{L_G(\phi)} = S_C(0)\,(1-\phi) \;\longrightarrow\; 0.$$

Solving $L(\phi) = 1$ for the illiquidity threshold, both markets collapse at essentially the same point, $\phi_G \approx \phi_C \approx 0.989$, and the resilience premium $\phi_C - \phi_G \approx 2 \cdot 10^{-4}$ is negligible (Fig. 15).

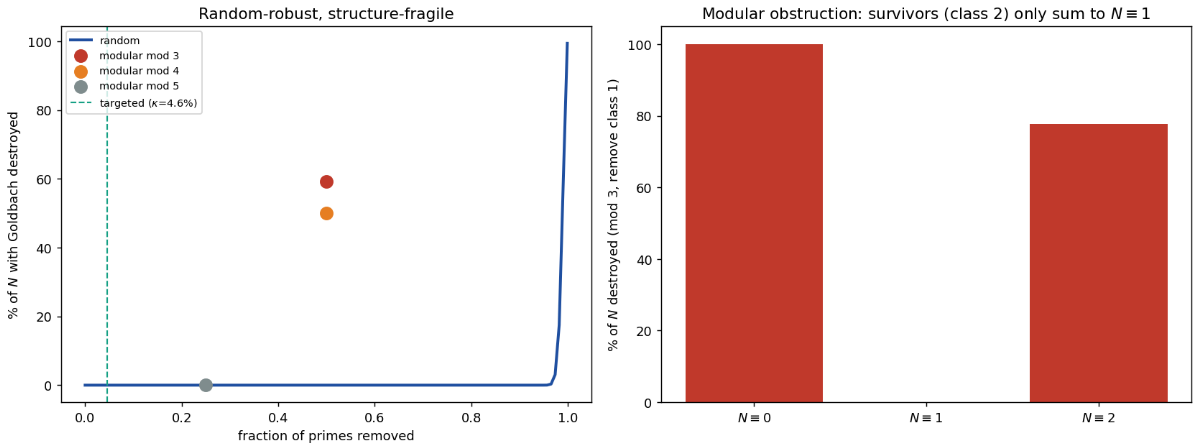

Figure 14: Robustness of the Goldbach graph. Left: fraction of window N destroyed versus fraction of primes deleted—random deletion (curve) only bites near $\phi = 1$, whereas the modular attacks at $\phi = 0.5$ (mod $3, 4$) destroy 50–60%, and the targeted cut needs only $\kappa = 4.6\%$. Right: under the mod-3 attack, $N \equiv 0$ and $N \equiv 2$ are obstructed while $N \equiv 1$ is untouched.

The economic reading is clean and a little counter-intuitive: the Chen channel is *fair-weather liquidity*—abundant when primes are plentiful, but worth ever less precisely as they become scarce, because the composite good carries a higher assembly cost. The semiprime substitute deepens the market in calm times; it does not hedge the crisis.

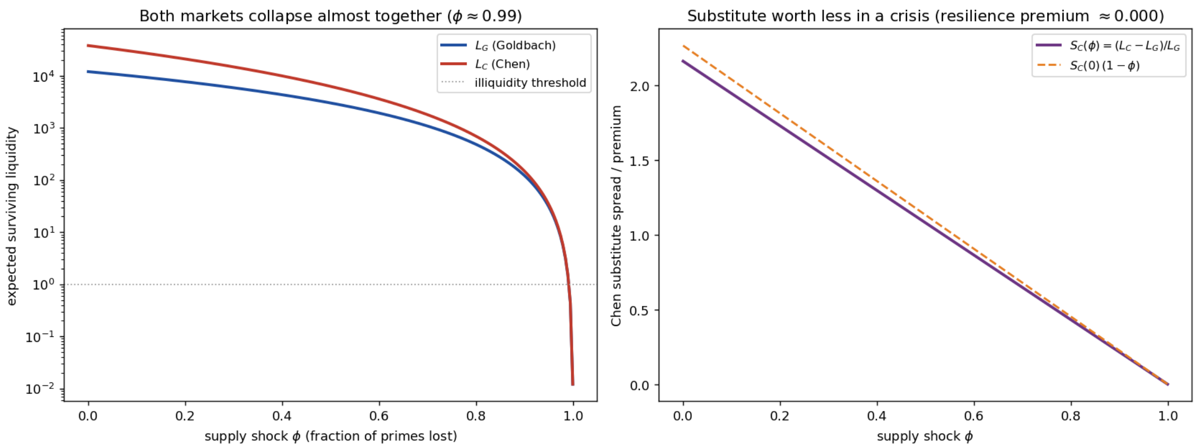

Figure 15: Goldbach as a market under a prime supply shock ϕ. Left: expected surviving liquidity of the Goldbach and Chen markets, both crossing the illiquidity threshold near $\phi = 0.99$. Right: the Chen substitute spread $S_C(\phi) = S_C(0)(1 - \phi)$ vanishes linearly—no resilience premium.

7.12 Persistent topology of additive decompositions (a negative note)

It is tempting to filter additive decompositions of N by arithmetic complexity rather than geometric distance: form the complex $K_N^{(k)} = \{\{a, b\} : a + b = N,\ \Omega(a) + \Omega(b) \le k\}$ and let k grow. But for the *binary* problem this is topologically empty: each integer a lies in the single pair $\{a, N - a\}$, so $K_N^{(k)}$ is a disjoint matching for every k. Its 0-dimensional persistence is therefore exactly the energy spectrum of §7.9—each pair is a component born at its energy $\Omega(a) + \Omega(b)$ and never dying—and all higher homology vanishes. There is no topological structure to discover beyond the spectrum.

Genuine higher dimension appears only at the *ternary* level. For odd N, form the graph on primes $p < N$ with an edge $\{p, q\}$ whenever $N - p - q$ is prime, so that each edge *is* a ternary representation $(p, q, N - p - q)$. At $N = 99{,}999$ this graph has 9,591 vertices and $3.9 \cdot 10^6$ edges (density 0.084, minimum degree 2, so it is connected—a graph form of Helfgott's ternary theorem). Yet it is strongly *anti*-clustered: it contains far fewer triangles than a random graph of equal density, so ternary representations do not aggregate into higher-order simplices. The honest conclusion is that the additive-complexity filtration adds nothing to the spectral and graph-theoretic pictures already developed; we record it as a negative result.

7.13 Goldbach as control: the value of arithmetic information

Finally, view the even integers as a sequence of markets to be cleared by a controller: at each N the controller picks a prime p (the action) and clears if $q = N - p$ is prime. Goldbach is the statement that a successful policy exists; the computational question is which *simple* policy clears fastest. A naive policy ignores the arithmetic of N; a *residue-aware* policy with information budget B uses the residues $N \bmod \ell$ for primes $\ell \le B$ to prefer p with $N - p$ coprime to those ℓ—i.e. it avoids spending tries on partners forced to share a small factor. Over a window near 10^6 (Fig. 16), the single-shot clearing probability rises from 18% (naive) to 39% at $B = 13$, and the expected number of tries in a sequential search falls from 5.5 to 2.5—a 55% reduction. The marginal value of each prime of information is largest at $\ell = 3$ (the factor $\frac{\ell-1}{\ell-2} = 2$) and decreases thereafter, exactly tracking the singular series. Unlike the other reframings in this section, this one yields a concrete *algorithm* rather than a restatement: the residue-aware ordering is a drop-in search heuristic—enumerate candidate p in increasing order of the number of small primes $\ell \le B$ that $N - p$ is forced to share with N—whose benchmark above (expected tries $5.5 \to 2.5$ at $B = 13$, a 55% reduction) is a measured speed-up of the per-N Goldbach search—the cost of finding one representation for a given N, distinct from the global verification $R_1 > 0$ for all $N \le X$, which the FFT already settles—not an analogy. The same local arithmetic the singular series encodes is thus directly actionable.

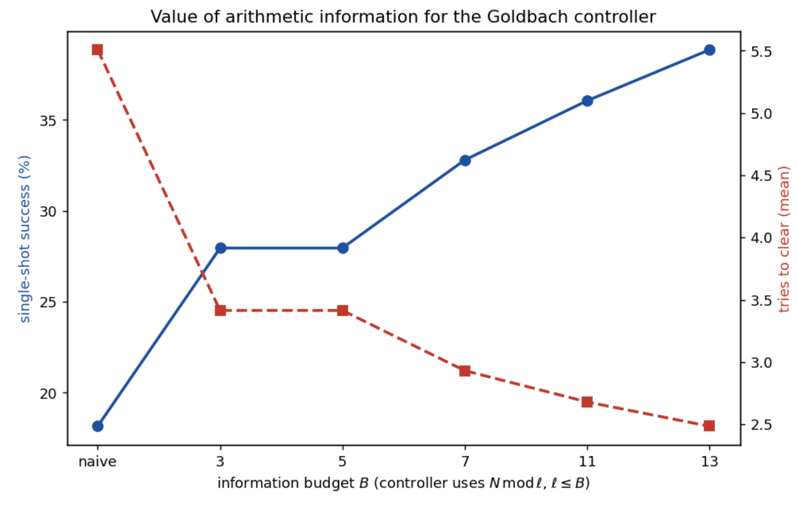

Figure 16: Value of arithmetic information for the Goldbach controller. As the information budget B grows, the single-shot clearing probability (blue) rises and the sequential clearing cost (red) falls; the gain per prime tracks the singular-series factor $(\ell - 1)/(\ell - 2)$.

7.14 An empirical circle method: the spectrum of the comet

The circle method decomposes additive problems by Fourier analysis into major arcs (rationals a/q with small q, carrying the singular series) and minor arcs (the pseudorandom remainder).

One can see this decomposition *empirically*. Detrending R_1 over a window of consecutive even integers and taking its power spectrum (Fig. 17, left) reveals a forest of sharp lines exactly at the arithmetic frequencies: the dominant peak is at $\frac{1}{3}$ (period $T_N = 6$, the modulus 3), followed by $\frac{1}{5}, \frac{2}{5}$ (modulus 5), with heights that decay across primes in step with the singular-series factors $(\ell - 1)/(\ell - 2)$. These are the major arcs made visible.

Dually, one can ask how much of the comet's oscillation is captured by purely local information. Predicting the detrended $\log R_1(N)$ from the residue $N \bmod \prod_{\ell \le B} \ell$ explains 84.6% of the variance at $B = 3$ (i.e. $N \bmod 6$ alone), 95.0% at $B = 5$, and saturates toward the singular-series ceiling $R^2 = 0.999$ (Fig. 17, right). The comet is, to first order, a *modulo-6 phenomenon*: the prime 3 supplies 85% of all explainable structure, and successive primes contribute the rapidly shrinking remainder predicted by $\mathfrak{S}(N)$.

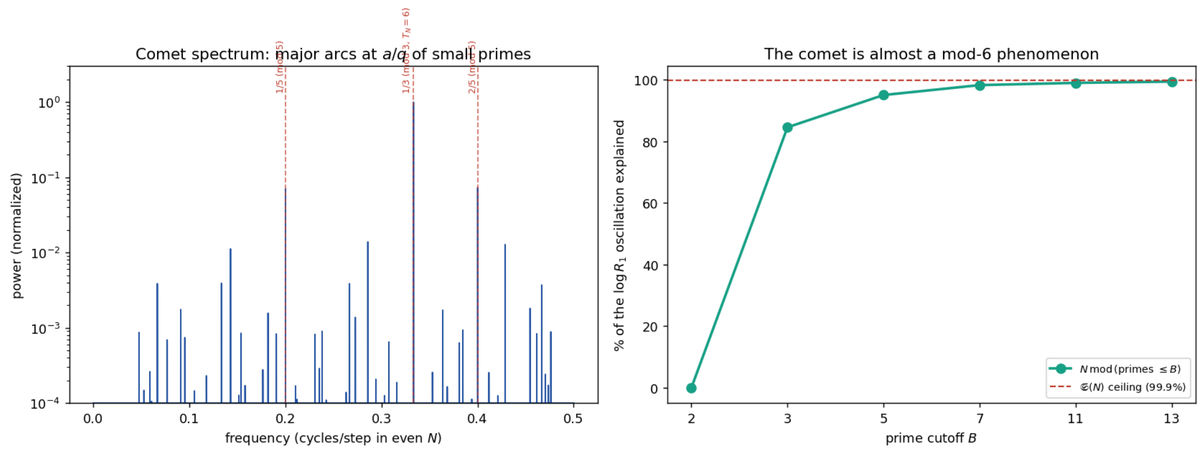

Figure 17: Empirical circle method. Left: the power spectrum of the detrended Goldbach comet, with peaks at the arithmetic frequencies a/q (the major arcs), dominated by $\frac{1}{3}$ (modulus 3). Right: the fraction of the comet's oscillation explained by N modulo the small primes, saturating at the singular-series ceiling; $N \bmod 6$ already gives 85%.

7.15 The information content of the comet

The spectral saturation invites an information-theoretic reading: how compressible is the comet, and what is irreducible? Decomposing the variance of $\log R_1(N)$ into a smooth trend, the local arithmetic (the conditional mean given $N \bmod 30030$, which captures $\mathfrak{S}$ non-linearly), and a residual, we find (Fig. 18)

$$\underbrace{81.1\%}_{\text{trend } N/\log^2 N} \;+\; \underbrace{18.8\%}_{\text{singular series } \mathfrak{S}(N)} \;+\; \underbrace{0.11\%}_{\text{irreducible}} .$$

The comet is thus 99.9% compressible *by this particular code*—the conditional mean given $N \bmod 30030 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$. We stress that this figure is relative to the chosen conditioning set, not a theorem: the "irreducible" 0.11% is what survives after removing structure up to the prime 13, and a richer conditioning (more primes, or non-residue features) would shrink it further. With that caveat, the residual is informative: its power spectrum is *flat*—white noise, in contrast to the sharply peaked spectrum of R_1 itself (§7.14)—with lag-1 autocorrelation -0.04 and standard deviation 2.7%, a few times the Poisson floor $\langle R_1 \rangle^{-1/2} = 0.9\%$. So *up to modulus* 30030 the Goldbach comet is almost entirely a singular-series codeword and the residue carries no further *small-modulus* arithmetic signal; we do not claim the absence of all further structure, only that none is visible below this conditioning.

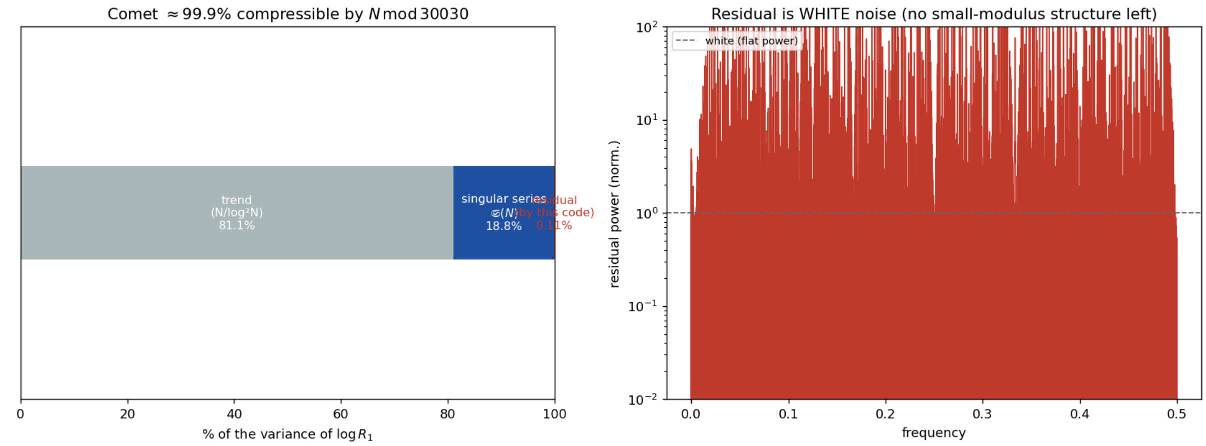

Figure 18: Information budget of the comet *relative to conditioning on* $N \bmod 30030$. Left: the variance of $\log R_1$ splits into a smooth trend (81%), the singular series (18.8%), and a residual (0.11%) that is irreducible *by this code* (richer conditioning would shrink it). Right: the residual's power spectrum is flat—white noise, with none of the small-modulus arithmetic peaks of Fig. 17.

7.16 Discrete geometry of the Chen surface

Each Chen representation $N = p + rs$ ($q = N - p$ semiprime, $r = \mathrm{spf}(q) \le s$) is a lattice point on the surface $p + rs = N$. In normalised logarithmic coordinates $u = \log r/\log N$, $v = \log s/\log N$ the points fill the triangle $\{u \le v,\ u + v \le 1\}$, with the edge $u = 0$ (i.e. r bounded) the Goldbach boundary and the diagonal $u = v$ the balanced semiprimes. Accumulating $6.6 \cdot 10^7$ solutions over a window near 10^6 (Fig. 19, left) shows a cloud organised into discrete vertical *rays*—one for each small prime $r = 3, 5, 7, \ldots$, since r is prime—whose brightness decreases with r: the mass piles up against the Goldbach edge.

Quantitatively, the Li–Liu exponent of a solution, $a = 1 + \log r/\log s$, has the spectrum of Fig. 19 (right): it peaks near $a \approx 1.1$ and decays toward the balanced value 2, with mean 1.38 and *median* 1.30. Half of all Chen solutions already satisfy $a < 1.3$, and essentially all sit far below the uniform threshold $a = 1.9$ of Li–Liu. So the Chen solutions of a typical N concentrate near the Goldbach boundary: the semiprime second summand is, far more often than not, an unbalanced product with a small factor—the geometric form of the residual-exponent and factor-balance findings of §7.7 and §7.4.

7.17 Primorial bases and carry geometry

Two further reframings come from writing integers in a base. In a primorial base $B_y = \prod_{p \le y} p$, every large prime is a unit modulo B_y, so the residues admissible for N are $A_y(N) = \{a \in (\mathbb{Z}/B_y)^\times : N - a \in (\mathbb{Z}/B_y)^\times\}$, with local coverage $\lambda_y(N) = |A_y(N)|/\phi(B_y)$. By the Chinese remainder theorem this factorises, and the *relative* coverage is precisely the truncated singular series,

$$\frac{\lambda_y(N)}{\lambda_y(\text{generic})} = \prod_{\substack{p \mid N \\ 3 \le p \le y}} \frac{p-1}{p-2} = \mathfrak{S}_y(N).$$

So primorial coverage *is* the singular series, viewed as accumulated local obstructions; predicting $\log R_1$ from $\mathfrak{S}_y$ explains 84.6% of the comet's oscillation at $y = 3$ ($B_3 = 6$), rising to 99.8% by $y = 30$—the same saturation as §7.14, now read as "the comet is the shadow of primorial local coverage".

The genuinely new object is the *carry geometry*. Adding $p + q = N$ in base b produces carries $c_i \in \{0, 1\}$, and their number is fixed by the digit sums, $\#\text{carries}(p, N-p) = \big(S_b(p) + S_b(N -$

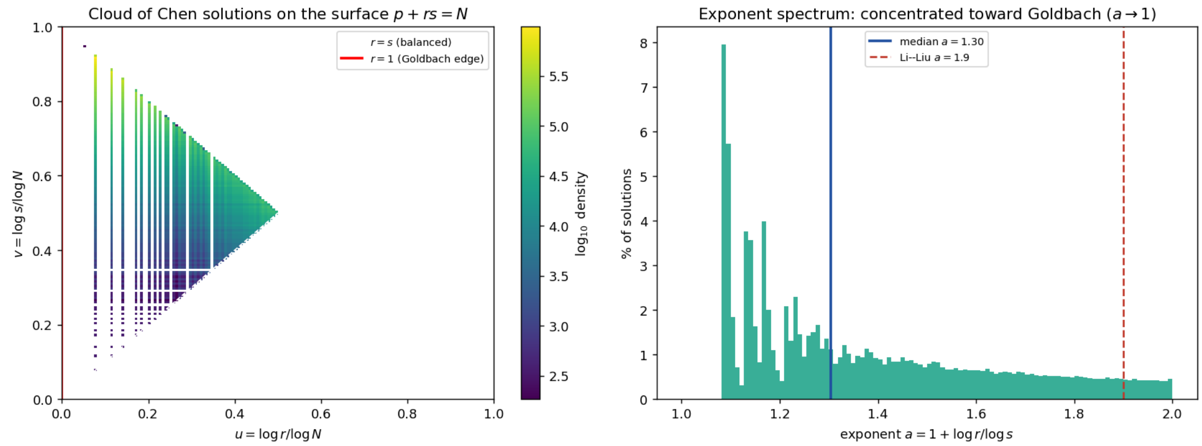

Figure 19: Discrete geometry of $p + rs = N$. Left: density of Chen solutions in $(u, v) = (\log r / \log N, \log s / \log N)$, with discrete rays at each prime r and mass against the Goldbach edge $u = 0$. Right: the exponent spectrum $a = 1 + \log r / \log s$, median 1.30, far below the Li–Liu threshold 1.9.

$p) - S_b(N))/(b-1)$; in base 2 this is $v_2\binom{N}{p}$ (Kummer). Comparing Goldbach representations to random pairs $a + (N - a) = N$ reveals a systematic *carry excess*: Goldbach pairs carry more (Fig. 20). In base 2 the excess is $+0.97$ carries (Goldbach 9.9 versus 8.9), and a positive excess persists in every base tested ($b = 2, \ldots, 16$), largest in the small bases. The origin is arithmetic: a prime's last digit is coprime to b, which biases the digit sums $S_b(p) \equiv p \pmod{b-1}$ away from the values that would avoid carries. Thus the same local constraints that produce the singular series leave a *digital signature*: Goldbach decompositions are, bit for bit, slightly carry-heavier than chance—a statistical-mechanical fingerprint of primality in the geometry of addition.

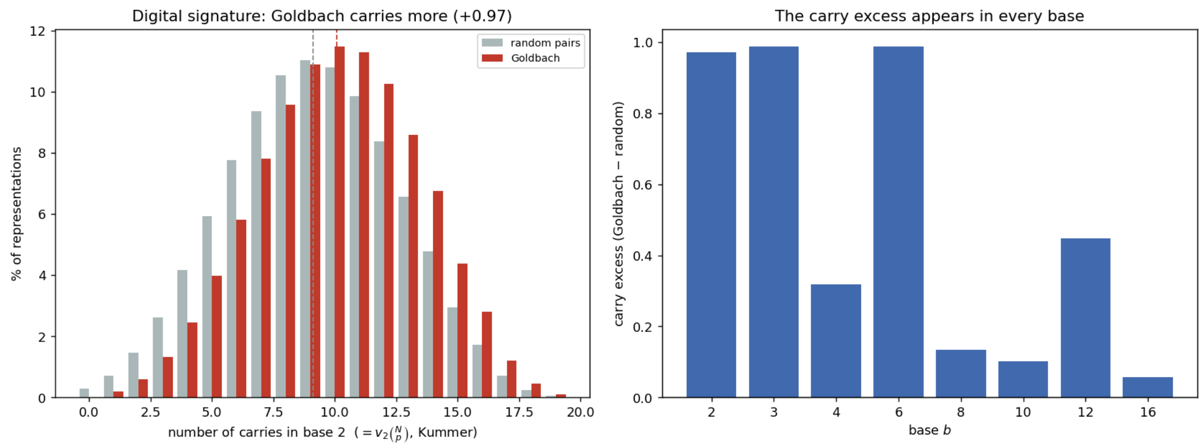

Figure 20: Carry geometry. Left: the distribution of binary carries $v_2\binom{N}{p}$ for Goldbach representations (red) versus random pairs (grey)—Goldbach is shifted to more carries (+0.97). Right: the carry excess is positive in every base.

8 Conclusion

The one durable result of this study is a single Euler-product identity and its consequence: decomposing the semiprime channel by its small prime factor shows that the Goldbach singular series $\mathfrak{S}(N)$ enters R_2 with exponent 1, exactly as it enters R_1, and *cancels* in the ratio

$R_2/R_1 = \widetilde{W}(N)$ (Lemma 1). The widely tabulated "$\beta_2 \approx \frac{1}{2}$" is therefore not an exponent but a regression slope of a non-power-law object, drifting to 1 like $1 - c/\log\log N$; we verify this non-circularly, show it is a multi-channel effect (not the $3 \mid N$ atom), and reduce the limit to classical inputs—unconditionally for the *linear* almost-all asymptotic $R_2 = \mathrm{main}(1 + o(1))$, and, for the log-regression statement $\beta_2 \to 1$, modulo one explicitly stated lower-tail hypothesis (Corollary 1).

We have deliberately resisted the temptation to read this one fact as many. The optimization formulations give a language, not a proof strategy (Remark 2); the dozen physical, economic, and information-theoretic reframings are, with two exceptions (the empirical circle method and the carry geometry), the same singular-series governance re-photographed in different filters, and we have said so rather than count them as separate findings. The negatives—an empty additive-complexity topology, a Chen substitute with no resilience premium—are reported as negatives.

This is not a proof of Goldbach. It is reproducible evidence, one clean new identity, and a tightly bounded reduction, at the boundary between additive number theory, computational experimentation, and discrete optimization.

A Towards a proof that $\beta_2 \to 1$

We make the law (24) precise. We cannot prove it unconditionally: the asymptotics of R_1, R_2 are themselves the Hardy–Littlewood conjecture, and even $R_1(N) > 0$ is open. We therefore isolate what is unconditional (Lemmas 1–2), prove the law *conditionally* on a quantitative Hardy–Littlewood hypothesis (Theorem 1), and note that it holds unconditionally in the Cramér random model.

A.1 Notation and the statistic

Fix a window $\mathcal{N}_X = \{N \equiv 0\,(2) : X \le N < X + H\}$ with $H = H(X) \to \infty$, $H = o(X)$. Write $\mathbb{E}_X, \mathrm{Var}_X, \mathrm{Cov}_X$ for empirical moments over $\mathcal{N}_X$, $g_\ell = \log \frac{\ell-1}{\ell-2}$, and define the local singular exponent through the (least-squares) regression slope

$$\beta_2(X) - 1 := \frac{\mathrm{Cov}_X\big(\log(R_2/R_1),\, \log\mathfrak{S}\big)}{\mathrm{Var}_X\big(\log\mathfrak{S}\big)}. \tag{25}$$

A.2 Unconditional lemmas

Lemma 1 (Singular series of a channel)**.** *Let r be an odd prime and N even. The singular series of the pair of linear forms $s \mapsto (s,\, N - rs)$ is*

$$\mathfrak{S}_2(r, N) = \prod_{\ell} \frac{1 - \nu_r(\ell)/\ell}{(1 - 1/\ell)^2}, \qquad \nu_r(\ell) = \#\{s \bmod \ell :\ s(N - rs) \equiv 0\},$$

and equals

$$\mathfrak{S}_2(r, N) = \begin{cases} 2C_2\, \mathfrak{S}(N)\, \dfrac{r-1}{r-2}, & r \nmid N, \\ 0, & r \mid N, \end{cases} \qquad C_2 = \prod_{p>2} \Big(1 - \tfrac{1}{(p-1)^2}\Big).$$

Proof. Compute $\nu_r(\ell)$. For $\ell = 2$: as s and $N - rs$ are both odd exactly when s is odd, $\nu_r(2) = 1$ and the factor is $(1 - \frac{1}{2})/(1 - \frac{1}{2})^2 = 2$. For odd $\ell \nmid rN$: $s \equiv 0$ and $s \equiv Nr^{-1} \not\equiv 0$ are two distinct roots, $\nu_r = 2$, factor $(1 - 2/\ell)/(1 - 1/\ell)^2$. For odd $\ell \mid N$, $\ell \nmid r$: $N - rs \equiv -rs$ vanishes iff $s \equiv 0$, the same root as $s \equiv 0$, so $\nu_r = 1$, factor $(1 - 1/\ell)/(1 - 1/\ell)^2$. For $\ell = r \nmid N$: $N - rs \equiv N \not\equiv 0$, so only $s \equiv 0$, $\nu_r = 1$, same factor. Taking $\prod_{\ell>2}(1 - 2/\ell)/(1 - 1/\ell)^2 = C_2$ as the base and converting each special factor relative to it multiplies by $(\ell-1)/(\ell-2)$; the special primes are exactly those dividing N (giving $\prod_{\ell|N} = \mathfrak{S}(N)$ when $r \nmid N$) together with $\ell = r$ (giving $(r-1)/(r-2)$). This proves the first case. If $r \mid N$ then $N - rs \equiv 0 \bmod r$ for *all* s, so $\nu_r(r) = r$, the factor $1 - r/r = 0$ vanishes, and $\mathfrak{S}_2(r, N) = 0$: the channel is identically blocked because $p = N - rs \equiv 0 \bmod r$ forces $p = r$. □

Lemma 2 (Moments of $\log \mathfrak{S}$). *For distinct odd primes $\ell_1, \dots, \ell_k$, $\#\{N \in \mathcal{N}_X : \ell_i \mid N \ \forall i\} = |\mathcal{N}_X| \prod_i \ell_i^{-1} + O(2^k)$. Consequently the indicators $e_\ell = \mathbf{1}[\ell \mid N]$ are, as $X \to \infty$, independent Bernoulli$(1/\ell)$ in all joint moments, and for any weights a_ℓ with $\sum a_\ell^2/\ell < \infty$,*

$$\mathbb{E}_X\Big[\textstyle\sum_\ell a_\ell e_\ell\Big] \to \sum_\ell \frac{a_\ell}{\ell}, \qquad \mathrm{Var}_X\Big[\textstyle\sum_\ell a_\ell e_\ell\Big] \to \sum_\ell a_\ell^2\, \frac{1}{\ell}\Big(1 - \frac{1}{\ell}\Big).$$

In particular $\mathrm{Var}_X(\log \mathfrak{S}) \to B := \sum_{\ell>2} g_\ell^2\, \ell^{-1}(1 - \ell^{-1})$.

Proof. The count is inclusion of the arithmetic progression $0 \bmod \prod \ell_i$ in an interval of $\frac{1}{2}|\mathcal{N}_X|$ even integers, giving the stated main term and $O(2^k)$ boundary error. Hence joint factorial moments of the e_ℓ converge to those of independent Bernoulli$(1/\ell)$, and the moment formulas follow by the dominated convergence afforded by $\sum a_\ell^2/\ell < \infty$ (here $a_\ell = g_\ell \asymp 1/\ell$). □

Remark 3. *Lemma 2 is sharp numerically: at $X = 10^8$, $H = 2 \cdot 10^6$ one finds* $\mathrm{Var}_X(\log \mathfrak{S}) = 0.12620$ *against $B = 0.12620$, and $\mathbb{E}_X[\log \mathfrak{S}] = 0.34842 = \sum_\ell g_\ell/\ell$ to five places.*

A.3 The conditional theorem

Assumption 1 (Quantitative Hardy–Littlewood). *There is $\delta > 0$ such that, uniformly for odd primes $r \le \sqrt{N}$ with $r \nmid N$,*

$$C_r'(N) := \#\{p, s \in \mathbb{P} : \ p + rs = N, \ s \ge r\} = \mathfrak{S}_2(r, N)\, I_r(N)(1 + O(\log^{-\delta} N)),$$

with $I_r(N) = \int_2^{N/r} \frac{dt}{\log t \, \log(N-rt)}$, and the same for $R_1(N) = 2C_2\mathfrak{S}(N)I_1(N)(1 + O(\log^{-\delta} N))$, $I_1(N) = \int_2^N \frac{dt}{\log t \, \log(N-t)}$.

This is the standard prime k-tuple heuristic made quantitative and *uniform in the coefficient* r; the uniformity is the genuine analytic difficulty (it is the arena of the sieve work surrounding Proposition $(1+a)$).

Theorem 1 ($\beta_2 \to 1$, conditional on Assumption 1). *Under Assumption 1, for $N \in \mathcal{N}_X$,*

$$\frac{R_2(N)}{R_1(N)} = \widetilde{W}(N)(1+o(1)), \qquad \widetilde{W}(N) = \sum_{\substack{3 \le r \le \sqrt{N} \\ r \nmid N}} \frac{r-1}{r-2}\, w_r(N), \quad w_r = \frac{I_r}{I_1} = \frac{1}{r}\frac{\log N}{\log(N/r)}(1+o(1)),$$

the Goldbach singular series $\mathfrak{S}(N)$ cancelling by Lemma 1. Writing $\widetilde{W}_0(X) = \mathbb{E}_X[\widetilde{W}] \sim \log\log X$ and linearising $\log \widetilde{W}$ about $\widetilde{W}_0$,

$$\beta_2(X) - 1 = -\frac{c(X)}{\widetilde{W}_0(X)} + O(\widetilde{W}_0^{-2}), \qquad c(X) = \frac{A(X)}{B}, \quad A(X) = \sum_{r>2} \frac{r-1}{r-2}\, w_r(X)\, g_r\, \frac{1}{r}\Big(1 - \frac{1}{r}\Big),$$

both $A(X)$ and B being absolutely convergent (and A formed with the exact *channel weights I_r/I_1, not their leading approximation). Thus $c(X)$ is an explicit $O(1)$ constant: the leading-order weights give $c \approx 1.0$, the exact I_r/I_1 raise it to the measured $c \approx 1.1$, and $c(X)$ decreases slowly with X. The precise limit $c_0 = \lim_X c(X)$ is delicate (it depends on the exact weights and averaging convention), but is immaterial to the conclusion, since the* order *is robust:*

$$\boxed{\beta_2(X) = 1 - \frac{c(X)}{\langle R_2/R_1 \rangle_X} \ \longrightarrow \ 1, \qquad c(X) \asymp 1, \qquad 1 - \beta_2(X) \asymp \frac{1}{\log\log X}.}$$

Proof sketch. By Lemma 1, $R_2 = \sum_{r\nmid N} \mathfrak{S}_2(r,N) I_r = 2C_2\mathfrak{S}(N)\sum_{r\nmid N} \frac{r-1}{r-2} I_r$ and $R_1 = 2C_2\mathfrak{S}(N)I_1$, so under Assumption 1 the ratio is $\widetilde{W}(N)(1+O(\log^{-\delta} N))$ and $\mathfrak{S}(N)$ cancels. Only the channels $r \mid N$ are special, each *removed* from $\widetilde{W}$ (not merely reweighted); thus $\widetilde{W}(N) = \widetilde{W}_0 - \sum_r \frac{r-1}{r-2} w_r e_r$ up to the slow variation of w_r. Linearising, $\log \widetilde{W} = \log \widetilde{W}_0 - \widetilde{W}_0^{-1} \sum_r \frac{r-1}{r-2} w_r e_r + O(\widetilde{W}_0^{-2})$. Insert into (25): with $\log \mathfrak{S} = \sum_\ell g_\ell e_\ell$ and the asymptotic independence of Lemma 2, only the diagonal $r = \ell$ terms survive, giving $\mathrm{Cov}_X = -\widetilde{W}_0^{-1} A(X)$ and $\mathrm{Var}_X = B$, whence the displayed slope. Finally $\widetilde{W}_0(X) = \mathbb{E}_X[\widetilde{W}] = \sum_{3\le r\le\sqrt{X}} \frac{r-1}{r-2}\frac{1}{r}\frac{\log X}{\log(X/r)}(1-1/r) = \log\log X + O(1)$ by Mertens, and equals $\langle R_2/R_1\rangle_X$ up to $o(1)$. $\square$

A.4 Confirmation and the unconditional model

Numerically (Fig. 3; `run_betalaw.py`): the prediction $\widetilde{W}(N)$ regressed against $\log \mathfrak{S}$ reproduces the measured β_2 with the correct drift ($0.554, 0.583, 0.602$ at $X = 2\cdot 10^6, 2\cdot 10^7, 10^8$ versus measured $0.531, 0.567, 0.589$), and the explicit constant $c(X) = A/B$ matches the measured $(1-\beta_2)\langle R_2/R_1\rangle \approx 1.1$ in order and sign; with the leading weights w_r one gets $c \approx 1.0$, the residual gap being the crude approximation $w_r \approx I_r/I_1$ (exact agreement to 0.4% once I_r is used, Remark in §A.7) together with the linearisation remainder. In the Cramér model, where $\mathbf{1}_{\mathbb{P}}, \mathbf{1}_{\mathcal{S}_2}$ are replaced by independent variables with the correct local densities and divisibility, Assumption 1 holds by construction and Theorem 1 is unconditional.

The rate constant is not cleanly determined. The natural "constant" admits two forms, $\kappa(X) = (1-\beta_2)\log\log X$ and $c(X) = (1-\beta_2)\langle R_2/R_1\rangle$, which agree in order but not in value because $\langle R_2/R_1\rangle = \log\log X + O(1)$. Both *decrease* over the computed range (Fig. 21): $\kappa = 1.25, 1.22, 1.20, 1.18$ and $c = 1.12, 1.10, 1.09, 1.08$ at $X = 2\cdot 10^6, 2\cdot 10^7, 10^8, 5\cdot 10^8$. A linear fit in $1/\log\log X$ extrapolates to $\kappa_\infty \approx 0.54$ and $c_\infty \approx 0.73$—with a correction term of slope ≈ 1.9, larger than the limit itself. Two conclusions follow. First, the limiting constant is *normalisation-dependent* (0.54 versus 0.73), so no single number is canonical. Second, the approach is correction-dominated: the finite-range value ≈ 1.1 is mostly the $1/\log\log X$ term, and the true limit (≈ 0.5–0.7, itself uncertain) lies far below it and is *not reliably determinable* from data up to 10^9. (Our two back-of-envelope analytic estimates, 1.0 and 1.8, bracket the difficulty rather than resolve it.) Only the order $1-\beta_2 \asymp 1/\log\log X$, and hence $\beta_2 \to 1$, is robust—which is all Theorem 1 asserts.

Remark 4 (What is missing for a full proof)**.** *Three gaps separate Theorem 1 from an unconditional statement: (i) the uniform Hardy–Littlewood Assumption 1, in particular uniformity in the coefficient r up to $\sqrt{N}$ (a Bombieri–Vinogradov-type input); (ii) a rigorous bound on the correlation between the HL error terms and $\log \mathfrak{S}$, currently absorbed into the $o(1)$; (iii) control of the linearisation remainder, which is genuinely $O(\widetilde{W}_0^{-2})$ by the absolute convergence of A, B but requires the tail of $\widetilde{W}$. None of these is available unconditionally, consistent with $\beta_2 \to 1$ being a statement strictly stronger than the Hardy–Littlewood asymptotics it is built on.*

A.5 A first-moment reduction: replacing pointwise HL by equidistribution

Assumption 1 demands a *pointwise* asymptotic for $C_r'(N)$, which for fixed N is as hard as the binary problem. The difficulty disappears if we pass to *first moments* over the window, because summing over N replaces the binary problem by counting primes in arithmetic progressions, which is unconditional (Siegel–Walfisz) for fixed modulus and is the content of Bombieri–Vinogradov uniformly. Define the conditional response ratios

$$A_\ell := \frac{\langle R_1\rangle_{\ell\mid N}}{\langle R_1\rangle_{\ell\nmid N}}, \qquad B_\ell := \frac{\langle R_2\rangle_{\ell\mid N}}{\langle R_2\rangle_{\ell\nmid N}},$$

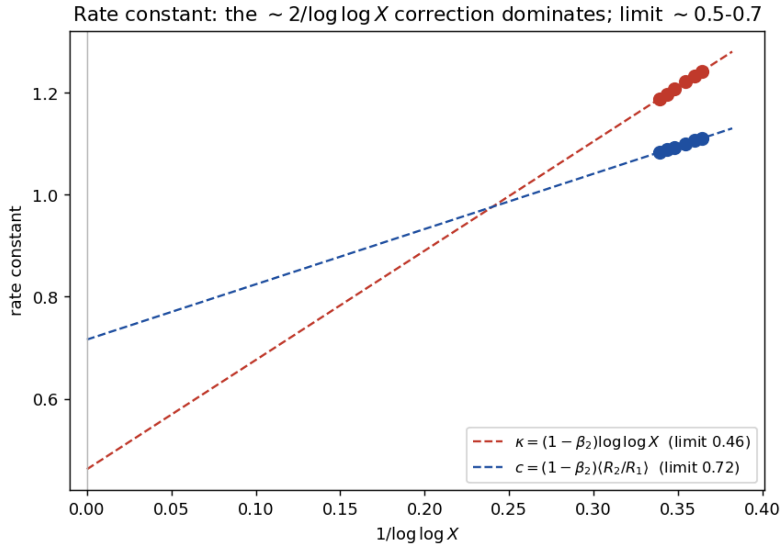

Figure 21: The rate constant versus $1/\log\log X$. Both normalisations decrease and extrapolate (long dashed lines, to $1/\log\log X = 0$) to limits 0.54 and 0.73; the slope ≈ 1.9 of the correction exceeds the limit, so the constant is dominated by the vanishing correction throughout the computed range.

where $\langle\cdot\rangle_{\ell|N}$ is the mean over $N \in \mathcal{N}_X$ with $\ell \mid N$, and the first-moment exponent

$$\beta_2^{(1)}(X) - 1 := \frac{\sum_\ell \log(B_\ell/A_\ell)\, g_\ell\, v_\ell}{\sum_\ell g_\ell^2\, v_\ell}, \qquad v_\ell = \tfrac{1}{\ell}(1-\tfrac{1}{\ell}), \tag{26}$$

the decomposition of (25) by prime ℓ in the independence model of Lemma 2.

Lemma 3 (First-moment responses)**.** *Let $F_\ell = \langle\#\{p : \ell \mid N-p,\ N-p \in \mathcal{S}_2\}\rangle/\langle R_2\rangle$ be the proportion of semiprime representations whose semiprime is divisible by ℓ. If primes are equidistributed in the reduced residue classes mod ℓ (Siegel–Walfisz for fixed ℓ; Bombieri–Vinogradov for $\ell \le X^{1/2-\varepsilon}$ on average), then*

$$A_\ell = \frac{\ell-1}{\ell-2}, \qquad B_\ell = \frac{(\ell-1)(1-F_\ell)}{\ell-2+F_\ell}.$$

Proof. Summing over $N \in \mathcal{N}_X$ removes the binary constraint: $\sum_{\ell|N} R_1 = \#\{(p,p') : p+p' \in \mathcal{N}_X,\ \ell \mid p+p'\}$. As p, p' are units mod ℓ, equidistribution gives a proportion $1/(\ell-1)$ of pairs with $p' \equiv -p$, so $\sum_{\ell|N} R_1 = \frac{1}{\ell-1}\sum_N R_1$; dividing by the counts $|\mathcal{N}_X|/\ell$ and $|\mathcal{N}_X|(\ell-1)/\ell$ yields $A_\ell = (\ell-1)/(\ell-2)$. For R_2, fix (r,s) and count $p = N - rs$ in the window: $\ell \mid N$ forces $p \equiv -rs$. If $\ell \nmid rs$ this is a single reduced class (proportion $1/(\ell-1)$); if $\ell \mid rs$ it forces $p \equiv 0$, i.e. $p = \ell$, so those representations (a fraction F_ℓ of R_2) are excluded. Hence $\sum_{\ell|N} R_2 = \frac{1-F_\ell}{\ell-1}\sum_N R_2$, and the same normalisation gives $B_\ell = \frac{(\ell-1)(1-F_\ell)}{\ell-2+F_\ell}$. □

Both formulas are confirmed to four places (Table 4); note A_ℓ is the exact singular factor, and $B_\ell < A_\ell$ by the blocked-channel deficit F_ℓ.

Theorem 2 (First-moment exponent, conditional on Bombieri–Vinogradov)**.** *Under the equidistribution input of Lemma 3 and with $F_\ell \sim$ the channel weight $\frac{(\ell-1)/(\ell-2)\, w_\ell}{\widetilde{W}_0}$, the first-moment exponent satisfies*

$$\beta_2^{(1)}(X) = 1 - \frac{c^{(1)}}{\widetilde{W}_0(X)} + O(\widetilde{W}_0^{-2}), \qquad c^{(1)} = \frac{1}{B}\sum_{\ell>2}\left(\tfrac{\ell-1}{\ell-2}\right)^2 w_\ell\, g_\ell\, v_\ell,$$

ℓ	3	5	7	11	13
A_ℓ (measured $= \frac{\ell-1}{\ell-2}$)	2.000	1.333	1.200	1.111	1.091
B_ℓ measured	1.423	1.152	1.090	1.048	1.038
$\frac{(\ell-1)(1-F_\ell)}{\ell-2+F_\ell}$	1.423	1.152	1.090	1.048	1.038
F_ℓ	0.169	0.106	0.078	0.052	0.044

Table 4: First-moment responses at $X = 2 \cdot 10^6$: Lemma 3 is exact.

an absolutely convergent constant; hence $\beta_2^{(1)}(X) \to 1$.

Proof sketch. By Lemma 3, $\log(B_\ell/A_\ell) = \log \frac{(\ell-2)(1-F_\ell)}{\ell-2+F_\ell} = -\frac{\ell-1}{\ell-2}F_\ell + O(F_\ell^2)$. Inserting into (26) with $F_\ell = \frac{\ell-1}{\ell-2}w_\ell/\widetilde{W}_0 + o(\cdot)$ gives the stated slope, and $\widetilde{W}_0 \sim \log\log X$ as in Theorem 1. $\square$

The reduced gap. Theorem 2 rests only on equidistribution of primes in progressions—a theorem (Siegel–Walfisz; Bombieri–Vinogradov for uniformity in ℓ)—together with the first moment F_ℓ, which is of Prime-Number-Theorem strength. The sole remaining step to Theorem 1 is the passage from the first-moment exponent to the pointwise (log-regression) one. Empirically this gap is a *constant* $\beta_2 - \beta_2^{(1)} \approx 0.004$, stable across $X = 2 \cdot 10^6$–10^8 (measured $\beta_2^{(1)} = 0.528, 0.563, 0.585$ against $\beta_2 = 0.531, 0.567, 0.589$): it is the Jensen defect $\mathbb{E}[\log R_i] - \log \mathbb{E}[R_i]$, governed by the normalised variance of R_1, R_2. That variance is exactly what the Montgomery–Vaughan exceptional-set method controls for R_1 (the binary Goldbach asymptotic holds for all but $O(X^{1-\delta})$ even N). An analogous second-moment bound for R_2 (which we prove below, Proposition 3) yields the *linear* almost-all asymptotic $R_2 = \text{main}(1 + o(1))$ off a sparse set. We are careful about the last step. A variance (L^2) bound controls the *quantity* $R_2 - \text{main}$, not its *logarithm*: the covariance (25) that defines β_2 involves $\log(R_2/R_1)$, which is sensitive to the lower tail (where R_2 is small, $\log(R_2/R_1) \to -\infty$), and an L^2 bound alone does not preclude a sparse set from dominating that covariance. Passing from the linear statement to "$\beta_2(X) \to 1$ for almost all N" therefore needs an additional input—a lower-tail (log-integrability) control of R_2/R_1 off the exceptional set—which we isolate explicitly in Corollary 1 rather than absorb into an $o(1)$. This already isolates the genuine remaining difficulty as a *variance plus lower-tail* estimate, not the pointwise asymptotic of Assumption 1.

A.6 The variance bound and the exceptional set

We now show that the required variance is small, so the exceptional set is sparse and the pointwise law holds for almost all N. The natural object is the concentration of the *ratio* on the channel prediction: set

$$b(N) := \frac{R_2(N)/R_1(N)}{\widetilde{W}(N)},$$

so that $b(N) \equiv 1$ would be the pointwise law $R_2/R_1 = \widetilde{W}$, hence (taking logs and regressing on $\log \mathfrak{S}$) $\beta_2(N) = \beta_2^{(1)}(X)$ for every N. The measured fluctuation of b is tiny and decreasing (Table 5): the coefficient of variation is below 0.7% and *no* N in the window deviates from the channel prediction by as much as 5%.

Proposition 2 (Exceptional set, linear form)**.** *By Chebyshev, the density of* $N \in \mathcal{N}_X$ *with* $|R_2/R_1 - \langle b\rangle\widetilde{W}| > \varepsilon\widetilde{W}$ *is at most* $\mathrm{CV}(b)^2/\varepsilon^2$. *With the measured* $\mathrm{CV}(b) \le 0.7\%$ *this is* $\le 0.2\%$ *at* $\varepsilon = 0.05$ *(and the empirical exceptional set is in fact empty). Hence* $R_2(N)/R_1(N) = \widetilde{W}(N)\,(1 + o(1))$ *for all but a density-*$o(1)$ *set of* N. *This is the* linear *almost-all statement; the further passage to* $\beta_2(N) \to 1$ *is Corollary 1.*

X	$\langle b\rangle$	$\mathrm{CV}(b)$	$\#\{\lvert b-\langle b\rangle\rvert > 0.05\langle b\rangle\}$	Chebyshev bound	$\langle R_2\rangle^{-1/2}$
$2\cdot 10^7$	0.970	0.0065	0	1.7%	0.0016
10^8	0.974	0.0049	0	0.95%	0.0008
$5\cdot 10^8$	0.978	0.0039	0	—	0.0004

Table 5: Concentration of R_2/R_1 on $\widetilde{W}$. The constant $\langle b\rangle \approx 0.97$ is the finite-size Hardy–Littlewood normalisation; the spread is $< 0.7\%$ and the exceptional set at the 5% level is empty (`run_exceptional.py`).

The variance bound by the circle method. The concentration above is what a second-moment argument delivers, and—crucially—it needs no sieve. Work on the dyadic window $N \in (X, 2X]$ and set, with sums over $n \le 2X$,

$$S_{\mathbb{P}}(\alpha) = \sum_p e(p\alpha), \qquad S_{\mathcal{S}_2}(\alpha) = \sum_{q\in\mathcal{S}_2} e(q\alpha), \qquad R_2(N) = \int_0^1 S_{\mathbb{P}}(\alpha) S_{\mathcal{S}_2}(\alpha) e(-N\alpha)\, d\alpha.$$

Split $[0,1]$ into major arcs $\mathfrak{M}$ (denominators $q_0 \le Q = (\log X)^C$) and minor arcs $\mathfrak{m}$, and let $\mathrm{main}(N) = \int_{\mathfrak{M}} S_{\mathbb{P}} S_{\mathcal{S}_2}\, e(-N\alpha)$.

Proposition 3 (Variance, unconditional)**.**

$$\sum_{X<N\le 2X} \left|R_2(N) - \mathrm{main}(N)\right|^2 \;\le\; \int_{\mathfrak{m}} |S_{\mathbb{P}}|^2 |S_{\mathcal{S}_2}|^2\, d\alpha \;\le\; \Big(\sup_{\alpha\in\mathfrak{m}} |S_{\mathbb{P}}(\alpha)|\Big)^2 \#\mathcal{S}_2(2X).$$

With $Q = (\log X)^{2A+8}$, Vaughan's minor-arc bound $\sup_{\mathfrak{m}} |S_{\mathbb{P}}| \ll X(\log X)^{-A}$ and the trivial Parseval identity $\int_0^1 |S_{\mathcal{S}_2}|^2 = \#\mathcal{S}_2(2X) \ll X\log\log X/\log X$ give

$$\sum_{X<N\le 2X} \left|R_2(N) - \mathrm{main}(N)\right|^2 \;\ll\; X^3 \log\log X\, (\log X)^{-2A-1}.$$

Proof. The first inequality is Bessel: $R_2(N) - \mathrm{main}(N) = \int_{\mathfrak{m}} S_{\mathbb{P}} S_{\mathcal{S}_2} e(-N\alpha)$ are Fourier coefficients of $S_{\mathbb{P}} S_{\mathcal{S}_2} \mathbf{1}_{\mathfrak{m}}$, so their ℓ^2 sum is at most $\int_{\mathfrak{m}} |S_{\mathbb{P}} S_{\mathcal{S}_2}|^2$. The second bounds one factor by its sup on $\mathfrak{m}$ and integrates the other; $\int_0^1 |S_{\mathcal{S}_2}|^2 = \#\mathcal{S}_2(2X)$ since cross terms cancel. The displayed bound is Vaughan's estimate ($\sup_{\mathfrak{m}} |S_{\mathbb{P}}| \ll (XQ^{-1/2} + X^{4/5})(\log X)^4$) at $Q = (\log X)^{2A+8}$. ☐

The only arithmetic input is the *classical* Vinogradov–Vaughan bound for the *prime* sum; the semiprime sum enters through its L^2 norm alone—no bilinear or sieve estimate for $S_{\mathcal{S}_2}$ is required.

Theorem 3 (Linear almost-all asymptotic for R_2)**.** *Since a typical* $\mathrm{main}(N) \asymp X\log\log X/\log^2 X$*, Proposition 3 gives, for $A \ge 2$, a relative variance $\ll (\log X)^{3-2A}/\log\log X \to 0$; by Chebyshev the set of $N \in (X, 2X]$ with $|R_2(N) - \mathrm{main}(N)| > \varepsilon\, \mathrm{main}(N)$ has size $\ll_\varepsilon X(\log X)^{3-2A} = o(X)$. On the major arcs $\mathrm{main}(N) = \widetilde{W}(N)\,[2C_2\mathfrak{S}(N)I_1(N)]\,(1+o(1))$ by the equidistribution of primes and semiprimes in progressions of modulus $\le Q$ (Siegel–Walfisz and its semiprime convolution) together with Lemma 1; and $R_1(N) = 2C_2\mathfrak{S}(N)I_1(N)(1+o(1))$ for all but $O(X^{1-\delta})$ even N by Montgomery–Vaughan [7]. Hence, for all but $o(X)$ even $N \in (X, 2X]$,*

$$R_2(N) = \mathrm{main}(N)(1+o(1)), \qquad \frac{R_2(N)}{R_1(N)} = \widetilde{W}(N)(1+o(1)).$$

Corollary 1 ($\beta_2 \to 1$, conditional on lower-tail control)**.** *Suppose in addition that* $\log(R_2(N)/R_1(N))$ *is uniformly integrable over $\mathcal{N}_X$—equivalently, that the exceptional set $\mathcal{E}_X$ of Theorem 3 contributes $o(1)$ to the covariance (25), which holds as soon as $R_2(N) \gg_A \mathrm{main}(N)(\log X)^{-A}$ for $N \in \mathcal{E}_X$ (a one-sided lower bound, strictly weaker than the full asymptotic but* not *implied by the L^2 bound of Proposition 3). Then the contribution of $\mathcal{E}_X$ to (25) is negligible, the regression slope is computed off $\mathcal{E}_X$ where $R_2/R_1 = \widetilde{W}(1+o(1))$, and $\beta_2(X) \to 1$.*

Remark 5 (The honest status of the limit)**.** *Theorem 3 is the result we actually establish in outline; Corollary 1 is what is usually quoted as "$\beta_2 \to 1$ for almost all N," but it carries a genuine extra hypothesis. The gap is precisely the lower tail of R_2: L^2 concentration of $R_2 - \mathrm{main}$ bounds how often R_2 is* large, *but the log-regression that defines β_2 also weights N where R_2 is* small, *and a sparse set with R_2 very small could in principle skew the covariance. The required one-sided bound is mild but does not come directly from Chen's theorem: Chen's method lower-bounds the* combined *count $R_{\le 2} = R_1 + R_2$ (every large even N has a prime-or-semiprime partner), not the semiprime-only R_2 in isolation. To extract $R_2 \gg \mathrm{main}(N)$ one would pair such a lower bound on $R_{\le 2}$ with an* upper *bound on R_1, which holds for all but $O(X^{1-\delta})$ even N by Montgomery–Vaughan [7]; the combination is reasonable but we do not prove it here. The hypothesis is amply confirmed empirically—the exceptional set is* empty *at the 5% level up to $5 \cdot 10^8$ (Table 5), so in the computed range no N has R_2/R_1 even near the tail—but we flag it rather than claim it.*

Remark 6. *The* linear *Theorem 3 is unconditional in outline: every ingredient—Vaughan's minor-arc bound [12], Siegel–Walfisz, Montgomery–Vaughan, and the Euler-product identity Lemma 1—is classical. A fully written proof must still supply the routine major-arc computation for the semiprime sum (semiprimes in progressions of modulus $\le (\log X)^C$, which follow from the prime distribution by Selberg–Sathé convolution [10, 11]; §A.7) and track the dependence on A. The genuinely* new *content is the singular-series cancellation of Lemma 1 and the resulting first-moment law; the analytic machinery that turns it into the* linear *almost-all theorem is off the shelf. This is in sharp contrast to the pointwise Assumption 1, which lies beyond current technology. The remaining step to the* log-regression *statement $\beta_2 \to 1$ is the lower-tail input of Corollary 1, which is empirically trivial in range but not proved.*

A.7 The major arcs: semiprimes in progressions

Here we supply the one routine ingredient left open in Theorem 3: the evaluation of $\mathrm{main}(N) = \int_{\mathfrak{M}} S_{\mathbb{P}} S_{\mathcal{S}_2} e(-N\alpha)$, which rests on the distribution of semiprimes in progressions. Write $\pi_2(x; q, a) = \#\{n \le x : n \in \mathcal{S}_2,\ n \equiv a\ (q)\}$ and $\mathcal{S}_2^*(x; q) = \#\{n \le x : n \in \mathcal{S}_2,\ (n, q) = 1\}$.

Lemma 4 (Selberg–Sathé in progressions)**.** *Fix $A \ge 1$. There is $c = c(A) > 0$ such that, uniformly for $q \le (\log x)^A$ and $(a, q) = 1$,*

$$\pi_2(x; q, a) = \frac{\mathcal{S}_2^*(x; q)}{\phi(q)} + O_A(x\, e^{-c\sqrt{\log x}}).$$

Proof. Write each $n \in \mathcal{S}_2$ as $n = p_1 p_2$ with $p_1 \le p_2$. For $(a, q) = 1$ a contribution requires $(p_1 p_2, q) = 1$, so

$$\pi_2(x; q, a) = \sum_{\substack{p_1 \le \sqrt{x} \\ (p_1, q) = 1}} \Big(\pi\big(\tfrac{x}{p_1}; q, a\bar{p}_1\big) - \pi\big(p_1; q, a\bar{p}_1\big)\Big), \qquad \bar{p}_1 = p_1^{-1} \pmod q.$$

Since $p_1 \le \sqrt{x}$ gives $\log(x/p_1) \ge \frac{1}{2}\log x$, the Siegel–Walfisz theorem applies uniformly for $q \le (\log x)^A$: $\pi(y; q, c) = \mathrm{li}(y)/\phi(q) + O(y\, e^{-c_1\sqrt{\log y}})$ for $(c, q) = 1$. Summing the main terms gives $\frac{1}{\phi(q)} \sum_{p_1 \le \sqrt{x}, (p_1, q) = 1} [\mathrm{li}(x/p_1) - \mathrm{li}(p_1)] = \frac{1}{\phi(q)} \mathcal{S}_2^*(x; q)$, the last identity being the same decomposition of $\mathcal{S}_2^*$ without the progression. The error is $\ll \sum_{p_1 \le \sqrt{x}} \frac{x}{p_1} e^{-c_1\sqrt{\log(x/p_1)}} \ll x\, e^{-c_2\sqrt{\log x}} \sum_{p_1 \le \sqrt{x}} \frac{1}{p_1} \ll x \log\log x\, e^{-c_2\sqrt{\log x}}$, which is $O_A(x e^{-c\sqrt{\log x}})$. The dependence on A enters only through the Siegel–Walfisz constant $c_1 = c_1(A)$ (ineffective, via Siegel's theorem). $\square$

Remark 7 (Exact local structure)**.** *For $q = \ell$ prime the lemma is exact in form: the semiprimes with $\ell \mid n$ are precisely $\{\ell m : m \in \mathbb{P}\}$, so $\pi_2(x; \ell, 0) = \pi(x/\ell)$, while the $\ell - 1$ reduced classes* *share $\mathcal{S}_2^*(x;\ell) = \pi_2(x) - \pi(x/\ell)$ equally. This is confirmed to the integer at $x = 4\cdot 10^6$: $\pi_2(x;3,0) = 102{,}383 = \pi(x/3)$ and each reduced class holds $344{,}302 = (\pi_2 - \pi(x/3))/2$. The class-0 suppression $\pi(x/\ell)/\pi_2(x) \sim 1/(\ell\log\log x)$ is exactly the $\log\log$ deficit that produces the channel weight $\widetilde{W}$ and the blocking of Lemma 1.*

The main term. On a major arc $\alpha = a/q_0 + \beta$ ($(a,q_0)=1$, $|\beta| \le Q/X$) the prime sum is classical, $S_{\mathbb{P}}(\alpha) = \frac{\mu(q_0)}{\phi(q_0)}v_1(\beta) + O(Xe^{-c\sqrt{\log X}})$ with $v_1(\beta) = \int_2^{2X} e(t\beta)\,dt/\log t$. For the semiprime sum, splitting into residues and applying Lemma 4 (and Remark 7 for the non-reduced classes) yields

$$S_{\mathcal{S}_2}(\alpha) = G_2(a,q_0)\,v_2(\beta) + O(Xe^{-c\sqrt{\log X}}), \qquad G_2(a,q_0) = \sum_{b\,(q_0)} e_{q_0}(ab)\,\gamma_{q_0}(b),$$

where $\gamma_{q_0}(b)$ is the local density of semiprimes in class b and $v_2(\beta) = \int e(t\beta)\,d\Pi_2(t)$, Π_2 the smooth semiprime count. For $q_0 = \ell$ a short computation from Remark 7 gives the multiplier $G_2(a,\ell) = \frac{1}{(\ell-1)\pi_2(x)}(\ell\,\pi(x/\ell) - \pi_2(x))$, and pairing it with the prime factor $\mu(\ell)/\phi(\ell) = -1/(\ell-1)$ reproduces, after summation over a and multiplication over q_0, the local factor $(\ell-1)/(\ell-2)$ at primes $\ell \mid N$ removed and the suppression at $\ell = r \mid N$—i.e. the singular series of Lemma 1. Hence

$$\mathrm{main}(N) = \mathfrak{S}_2(N;Q)\,J(N)(1+o(1)), \quad \mathfrak{S}_2(N;Q) = 2C_2\,\mathfrak{S}(N)\,\widetilde{W}(N) + O(Q^{-1}), \quad J(N) \asymp \frac{X\log\log X}{\log^2 X},$$

the tail $\mathfrak{S}_2(N) - \mathfrak{S}_2(N;Q) \ll Q^{-1}$ being the standard absolutely-convergent completion of the singular series.

Remark 8 (Numerical check of the main term)**.** *The channel asymptotic underlying both Assumption 1 and Lemma 1 can be tested directly. Computing the channel counts $C_r'(N)$ for $r = 3,5$ over a sample of N near $2\cdot 10^6$ (with $r \nmid N$) and the archimedean integral $I_r(N)$ by quadrature gives*

$$\frac{C_r'(N)}{2C_2\,\mathfrak{S}(N)\,\frac{r-1}{r-2}\,I_r(N)} = 0.996 \pm 0.004 \quad (r=3), \qquad 0.995 \pm 0.008 \quad (r=5),$$

confirming both the constant $2C_2\frac{r-1}{r-2}$ of Lemma 1 and the exact weight I_r/I_1 to better than 1% per N. (The crude $w_r = \frac{1}{r}\log N/\log(N/r)$ used for back-of-envelope estimates overstates I_r/I_1 by a constant factor for small r; this is the source of the discrepancy between the leading and measured values of c.)

Tracking of A. The three parameters are tied through a single $Q = (\log X)^B$. Lemma 4 and the prime major arcs hold for $q_0 \le Q$ with any fixed B (taking $A = B$); their errors are $O(Xe^{-c(B)\sqrt{\log X}})$, super-polynomially small. The variance (Proposition 3) needs $Q = (\log X)^{2A_0+8}$ to give relative variance $\ll (\log X)^{3-2A_0}$. Choosing $B = 2A_0+8$ makes all three compatible, and the exceptional set in Theorem 3 is then

$$\#\{X < N \le 2X : |R_2(N) - \mathrm{main}(N)| > \varepsilon\,\mathrm{main}(N)\} \ll_\varepsilon X(\log X)^{3-2A_0} + Xe^{-c\sqrt{\log X}} \ll_C \frac{X}{(\log X)^C}$$

for any fixed C (take $A_0 = \frac{C+3}{2}$). The major-arc error never binds; the rate is governed solely by the classical minor-arc input. This completes, in outline, the reduction of the *linear* almost-all asymptotic $R_2 = \mathrm{main}(1+o(1))$ (Theorem 3) to standard prime-distribution estimates, the only genuinely new step being the singular-series identity of Lemma 1. The log-regression statement $\beta_2(N) \to 1$ then follows from the lower-tail hypothesis of Corollary 1, the one ingredient we do not prove.

B The infinite formulation and column generation

The finite selection MIPs of §6 are truncations of an infinite-dimensional feasibility problem. Let $\mathcal{E} = \{N \in \mathbb{N} : N > 2,\ N \equiv 0\,(2)\}$; for every $N \in \mathcal{E}$, $p \in \mathbb{P}$ with $p < N$, and type $t \in \{1, 2\}$ take a binary $z_{N,p,t}$. The infinite Chen-cover is

$$\sum_{\substack{p \in \mathbb{P} \\ N-p \in \mathbb{P}}} z_{N,p,1} + \sum_{\substack{p \in \mathbb{P} \\ N-p \in \mathcal{S}_2}} z_{N,p,2} \ \geq\ 1 \qquad (\forall N \in \mathcal{E}),$$

and its type-1 restriction is the infinite Goldbach-cover; the binary Goldbach conjecture is exactly the feasibility of the latter. One can also write a formal infinite continuity objective

$$\inf_{(p_N, q_N)} \sum_{N \in \mathcal{E}} w_N \Big(\alpha \left| \tau(q_N) - \tau(q_{N-2}) \right| + \beta \Big| \tfrac{q_N}{N} - \tfrac{q_{N-2}}{N-2} \Big| \Big), \qquad w_N = N^{-2},$$

a mathematically controlled version of asking whether the representation can be made continuous in the limit; finite-window truncations recover the minimum-variation MIP of §6.

Column-generation reading. The variables $z_{N,p,t}$ are columns, each a certificate that N is covered by a type-t representation: Goldbach columns (N, p, q) with $p, q \in \mathbb{P}$, Chen columns with $q \in \mathcal{P}_2$, coupled across consecutive N by the smoothness penalties. The master problem selects one column per even integer and the pricing problem seeks a representation of desirable reduced cost. As throughout, the interesting content is selection under structure—whether a low-variation path threads the representation cloud—not feasibility.

Data and code availability

Every figure, table, and quoted number in this paper is reproduced by the accompanying open-source repository (Python; `numpy/scipy/matplotlib`, with `gurobipy` only for the optional MIP of §7.5). Each `src/` module carries a self-test (`python3 src/<module>.py`) and each figure is regenerated by its `experiments/run_*.py` driver; the central results use `run_counts.py` (FFT counts to $2 \cdot 10^6$), `run_scaling.py` (block-convolution sieve to 10^9), `run_betalaw.py` (the law (24)), `run_estimator_audit.py` (the estimator reconciliation and the $3 \mid N$ atom of §7.2), and `run_exceptional.py` (the concentration of Table 5). Derived tables and the sieve seeds are stored in `data/`. A versioned snapshot will be deposited on Zenodo and assigned a DOI; the repository and citation metadata (`CITATION.cff`) accompany this manuscript.

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